

Fundamentals of Physics

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Vectors

1.1 Introduction to Scalars and Vectors

Let us begin with a piece of arithmetic that looks broken. In this chapter, $4 + 3$ will sometimes equal 7. It will sometimes equal 5. It can equal 1, and with the right setup it can equal anything in between. By the end of the chapter you will not only see why, you will be able to compute exactly what $4 + 3$ equals in any given situation — and you will have met the single most important mathematical tool of this whole course.

Here is the puzzle in concrete form. Walk 4 m, stop, then walk 3 m more. How far are you from where you started?

You cannot answer — and that is the point. It depends on which *direction* the second walk took. Walk 4 m east then 3 m east: you are 7 m from the start. Walk 4 m east then 3 m *west*: you are 1 m from the start. And if you walk 4 m east, turn a corner, and walk 3 m north:

- The total *distance* you walked is $4 + 3 = 7$ m. Distance only cares about how much, not which way.
- Your *displacement* — the straight-line shortcut from start to finish — is only 5 m (a 3-4-5 right triangle), pointing diagonally. Displacement cares about how much *and* which way.

So there are two kinds of quantities in physics. Some, like distance, mass, and temperature, are single numbers and obey ordinary arithmetic. Others, like displacement, carry a direction with them, and adding them requires new rules — rules in which $4 + 3$ can come out anywhere from 1 to 7. Quantities of the first kind are called *scalars*; quantities of the second kind, *vectors*.

Scalars and Vectors

A **scalar** is a quantity that has magnitude only (no direction): distance, mass, time, temperature, speed, energy.

A **vector** is a quantity that has magnitude and direction: displacement, velocity, acceleration, force.

	Scalar	Vector
Has magnitude	yes	yes
Has direction	no	yes
Examples	distance, speed, mass, time	displacement, velocity, force

Notation

- Vectors are written with a small arrow above the symbol, e.g. a force $\vec{F}$ or a velocity $\vec{v}$.
- The magnitude (size) of a vector $\vec{A}$ is written $|\vec{A}|$, or simply A .
- In diagrams a vector is drawn as an arrow: the length of the arrow is proportional to the magnitude, and the arrowhead shows the direction.

Vectors have magnitude and direction, but not location

A displacement of 1 m due east from Paro is represented by the *same* vector as a displacement of 1 m due east from Thimphu. Sliding a vector to a new position (without rotating or resizing it) does not change the vector at all.

The negative of a vector

$-\vec{A}$ is a vector with the same magnitude as $\vec{A}$ but pointing in the opposite direction.

1.2 Vector Operations

How do forces add? Nobody knew for two thousand years

It is tempting to think the rules below were always obvious. They were not. That two forces acting on the same object combine like two walks — direction and all — is a *discovery about nature*, and it took until 1586. That year a Flemish engineer named Simon Stevin published *De Beghinselen der Weeghconst* (“The Principles of the Art of Weighing”), in which he worked out how weights pull on objects resting on inclined planes. His famous argument used an endless chain of balls draped over a triangular wedge: since the chain cannot spin around the wedge forever on its own (that would be a perpetual motion machine, which Stevin refused to believe in), the pulls on the two slopes must exactly balance — and from this he could deduce how a force splits along and across a slope. Stevin was so proud of the argument that he printed the picture on the title page of his book with the motto “*Wonder en is gheen wonder*” — “what seems a wonder is no wonder.” That motto could be the slogan for this entire course.

Out of this line of work — Stevin, then Galileo, then Newton — came the rule that forces, velocities, and displacements all combine the same way: like arrows. Here is that rule.

Addition of vectors (head-to-tail / triangle rule)

To add two vectors, place the tail of $\vec{B}$ at the head of $\vec{A}$. The sum $\vec{A} + \vec{B}$ is the vector drawn from the tail of $\vec{A}$ to the head of $\vec{B}$. This sum is called the **resultant vector**.

Addition is **commutative**: $\vec{A} + \vec{B} = \vec{B} + \vec{A}$.

Addition is **associative**: $(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$.

Subtraction means adding the opposite: $\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$.

On the companion site: open the **vector sandbox** (Chapter 1 on the course website) and drag two arrows around. Watch the head-to-tail sum update live — then try to make $4 + 3$ equal exactly 5.

Parallelogram law of vector addition

Draw $\vec{a}$ and $\vec{b}$ with both tails together. Treat them as two adjacent sides and complete the parallelogram. The diagonal through the common tails is the sum $\vec{a} + \vec{b} = \vec{R}$.

Stop and Predict 1.1: Taming $4 + 3$

Two tug ropes pull on the same hook: one with 4 N, one with 3 N. Before any formula, write down your answers:

1. What is the *largest* the combined pull could possibly be, and how must the ropes be arranged to get it?
2. What is the *smallest*, and how?
3. At what angle (roughly) between the ropes would the combined pull be exactly 5 N?

Keep your three guesses — the formula we are about to derive will grade them.

Deriving the magnitude and direction of the resultant. Let $a = |\vec{a}|$, $b = |\vec{b}|$, and let θ be the angle between them. By the law of cosines applied to the triangle formed:

$$R^2 = a^2 + 2ab \cos \theta + b^2,$$

so

Magnitude and direction of the resultant

$$R = |\vec{a} + \vec{b}| = \sqrt{a^2 + 2ab \cos \theta + b^2}$$
$$\alpha = \tan^{-1} \left(\frac{b \sin \theta}{a + b \cos \theta} \right) \quad (\text{angle that } \vec{R} \text{ makes with } \vec{a})$$

Worked Example 1.1: Resultant of two forces

Two forces of $a = 3$ N and $b = 4$ N act at a point with an angle of $\theta = 60^\circ$ between them. Find the magnitude and direction of the resultant.

Solution.

$$R = \sqrt{3^2 + 2(3)(4) \cos 60^\circ + 4^2} = \sqrt{9 + 12 + 16} = \sqrt{37} \approx 6.1 \text{ N.}$$

$$\alpha = \tan^{-1} \left(\frac{4 \sin 60^\circ}{3 + 4 \cos 60^\circ} \right) = \tan^{-1} \left(\frac{3.46}{5} \right) \approx 34.7^\circ$$

measured from the 3 N force.

Now grade your predictions from Stop and Predict 1.1. Setting $\theta = 0^\circ$ (ropes parallel) gives $R = \sqrt{9 + 24 + 16} = 7 \text{ N}$ — the largest possible. Setting $\theta = 180^\circ$ (ropes opposite) gives $R = \sqrt{9 - 24 + 16} = 1 \text{ N}$ — the smallest. And $R = 5 \text{ N}$ requires $25 = 25 + 24 \cos \theta$, i.e. $\cos \theta = 0$: the ropes must be exactly *perpendicular*. Our broken arithmetic is fixed: $4 + 3$ equals anything from 1 to 7, and the angle decides.

Checkpoint 1.1: Two equal pulls

Two students pull on a ring, each with a force of 100 N, with an angle of 120° between their ropes. The resultant force on the ring has magnitude

- (a) 200 N
- (b) 173 N
- (c) 141 N
- (d) 100 N

Commit to an answer before checking — and try to feel why the popular wrong answers are tempting.

Answer: (d). With $a = b = 100$ and $\theta = 120^\circ$, $R = \sqrt{100^2 + 2(100)(100)(-\frac{1}{2})} = 100$. The two pulls and the resultant form an equilateral triangle — each pull “wastes” half of itself fighting the other.

Try It Yourself 1.1: Two velocities

A boat heads north at $a = 6 \text{ m/s}$ while a current pushes it at $b = 8 \text{ m/s}$ at $\theta = 90^\circ$ to the boat’s heading. Show that the resultant speed is 10 m/s and find its direction relative to north.

Multiplication by a scalar

Multiplying a vector by a positive scalar n multiplies its magnitude but leaves the direction unchanged. If n is negative, the direction is reversed. Scalar multiplication is distributive:

$$n(\vec{A} + \vec{B}) = n\vec{A} + n\vec{B}.$$

The dot (scalar) product

Dot product

$$\vec{A} \cdot \vec{B} \equiv AB \cos \theta,$$

where θ is the angle between the vectors when placed tail to tail. The result is a scalar — hence the alternative name *scalar product*.

Geometrically, $\vec{A} \cdot \vec{B}$ is the magnitude of $\vec{A}$ times the projection of $\vec{B}$ along $\vec{A}$ (which is $B \cos \theta$), or equivalently B times the projection of $\vec{A}$ along $\vec{B}$.

Properties.

- Commutative: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$.
- Distributive: $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$.
- A vector dotted with itself gives its magnitude squared: $\vec{A} \cdot \vec{A} = A^2$.
- If $\vec{A} \perp \vec{B}$, then $\vec{A} \cdot \vec{B} = AB \cos 90^\circ = 0$.

Worked Example 1.2: Work done by a force

A force $\vec{F}$ of magnitude 10 N acts at 30° to a displacement $\vec{d}$ of magnitude 5 m. The work done is $W = \vec{F} \cdot \vec{d}$:

$$W = Fd \cos \theta = (10)(5) \cos 30^\circ \approx 43.3 \text{ J}.$$

Worked Example 1.3: Class exercise — the Law of Cosines

Let $\vec{C} = \vec{A} - \vec{B}$. Calculate the dot product of $\vec{C}$ with itself.

Solution.

$$\begin{aligned}\vec{C} \cdot \vec{C} &= (\vec{A} - \vec{B}) \cdot (\vec{A} - \vec{B}) = \vec{A} \cdot \vec{A} - \vec{A} \cdot \vec{B} - \vec{B} \cdot \vec{A} + \vec{B} \cdot \vec{B} \\ &= A^2 - AB \cos \theta - BA \cos \theta + B^2.\end{aligned}$$

$$\boxed{C^2 = A^2 + B^2 - 2AB \cos \theta} \quad (\text{the Law of Cosines}).$$

The cross (vector) product

Cross product

$$\vec{A} \times \vec{B} \equiv AB \sin \theta \hat{n},$$

where $\hat{n}$ is a unit vector pointing perpendicular to the plane containing $\vec{A}$ and $\vec{B}$. The result is itself a vector — hence the name *vector product*.

The magnitude is $|\vec{A} \times \vec{B}| = AB \sin \theta$.

Right-hand rule

Point the fingers of your right hand along the first vector, curl them through the smaller angle toward the second vector — your thumb then points along $\hat{n}$.

Properties.

- Distributive: $\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$.
- **Not commutative** — order matters: $\vec{B} \times \vec{A} = -(\vec{A} \times \vec{B})$.
- Geometrically, $|\vec{A} \times \vec{B}|$ equals the area of the parallelogram with sides $\vec{A}$ and $\vec{B}$.
- Parallel vectors give zero: $\vec{A} \times \vec{A} = A^2 \sin 0^\circ \hat{n} = \vec{0}$.

Worked Example 1.4: Torque

A spanner is turned by a force $F = 20 \text{ N}$ applied at the end of a handle of length $r = 0.30 \text{ m}$, at 90° to the handle. The torque magnitude is

$$|\vec{\tau}| = |\vec{r} \times \vec{F}| = rF \sin 90^\circ = (0.30)(20)(1) = 6 \text{ N m}.$$

1.3 Components of a Vector

It is often useful to describe a vector in terms of its components. If a vector $\vec{A}$ is split into two mutually perpendicular vectors, these are called the **rectangular components** of $\vec{A}$.

The vector $\vec{A}$ has magnitude A and makes angle θ with the positive x -axis.

Finding the components from magnitude and angle

$$A_x = A \cos \theta, \quad A_y = A \sin \theta.$$

We can reverse the process to recover the magnitude and direction:

$$A_x^2 + A_y^2 = A^2(\cos^2 \theta + \sin^2 \theta) = A^2, \quad \frac{A_y}{A_x} = \tan \theta.$$

Finding magnitude and angle from the components

$$A = \sqrt{A_x^2 + A_y^2}, \quad \theta = \tan^{-1}\left(\frac{A_y}{A_x}\right).$$

Worked Example 1.5: Resolving a force

A force of $A = 100 \text{ N}$ acts at $\theta = 30^\circ$ above the horizontal. Its components are

$$A_x = 100 \cos 30^\circ \approx 86.6 \text{ N}, \quad A_y = 100 \sin 30^\circ = 50 \text{ N}.$$

Worked Example 1.6: Adding vectors using components

Add $\vec{A}$ (5 m at 37°) and $\vec{B}$ (8 m at 0° , i.e. along x).

Solution. Resolve each, add like components, then recombine.

$$\begin{aligned}A_x &= 5 \cos 37^\circ \approx 4.0, & A_y &= 5 \sin 37^\circ \approx 3.0, \\B_x &= 8, & B_y &= 0.\end{aligned}$$

$$R_x = 4.0 + 8 = 12.0, \quad R_y = 3.0 + 0 = 3.0.$$

$$R = \sqrt{12.0^2 + 3.0^2} \approx 12.4 \text{ m}, \quad \theta = \tan^{-1}\left(\frac{3.0}{12.0}\right) \approx 14^\circ.$$

Try It Yourself 1.2: Components of velocity

A projectile is launched at 20 m/s, 60° above the horizontal. Find the horizontal and vertical components of the launch velocity.

(Answer: $v_x = 10 \text{ m/s}$, $v_y \approx 17.3 \text{ m/s}$.)

1.4 Vector Algebra in Components

In practice it is often easiest to set up Cartesian coordinates x, y, z and work with components.

Basis vectors and unit vectors

Let $\hat{x}, \hat{y}, \hat{z}$ (sometimes written $\hat{i}, \hat{j}, \hat{k}$) be unit vectors pointing along the x -, y -, and z -axes. These are the **basis vectors**.

Unit vector

A unit vector has magnitude exactly 1; its only job is to indicate direction. Any vector $\vec{A}$ can be turned into a unit vector (written $\hat{A}$) by dividing by its own magnitude:

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}.$$

Component form of a vector

Any vector can be written in terms of the basis vectors:

$$\vec{A} = A_x \hat{x} + A_y \hat{y} + A_z \hat{z}.$$

The coefficients A_x, A_y, A_z are the *components* of $\vec{A}$ — geometrically, the projections of $\vec{A}$ onto the three axes. A component can be extracted by dotting with the matching basis vector:

$$\vec{A} \cdot \hat{x} = A_x, \quad \vec{A} \cdot \hat{y} = A_y, \quad \vec{A} \cdot \hat{z} = A_z.$$

The four operations in component form

Rule 1 — Addition: add like components

$$\vec{A} + \vec{B} = (A_x + B_x)\hat{x} + (A_y + B_y)\hat{y} + (A_z + B_z)\hat{z}$$

Rule 2 — Scalar multiplication: scale every component

$$a\vec{A} = (aA_x)\hat{x} + (aA_y)\hat{y} + (aA_z)\hat{z}$$

Rule 3 — Dot product: multiply like components and add

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

This works because the basis vectors are mutually perpendicular unit vectors:

$$\hat{x} \cdot \hat{x} = \hat{y} \cdot \hat{y} = \hat{z} \cdot \hat{z} = 1, \quad \hat{x} \cdot \hat{y} = \hat{x} \cdot \hat{z} = \hat{y} \cdot \hat{z} = 0.$$

In particular $\vec{A} \cdot \vec{A} = A_x^2 + A_y^2 + A_z^2$, so

Magnitude (3-D Pythagoras)

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

Bonus — Rule 4: the cross product as a determinant

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = (A_y B_z - A_z B_y)\hat{x} - (A_x B_z - A_z B_x)\hat{y} + (A_x B_y - A_y B_x)\hat{z}$$

Worked Example 1.7: Putting the rules together

Let $\vec{A} = 2\hat{x} + 3\hat{y} + \hat{z}$ and $\vec{B} = \hat{x} - \hat{y} + 4\hat{z}$.

(a) *Sum.* $\vec{A} + \vec{B} = (2 + 1)\hat{x} + (3 - 1)\hat{y} + (1 + 4)\hat{z} = 3\hat{x} + 2\hat{y} + 5\hat{z}$.

(b) *Dot product.* $\vec{A} \cdot \vec{B} = (2)(1) + (3)(-1) + (1)(4) = 2 - 3 + 4 = 3$.

(c) *Magnitude of $\vec{A}$.* $|\vec{A}| = \sqrt{2^2 + 3^2 + 1^2} = \sqrt{14} \approx 3.74$.

(d) *Angle between them.* Using $\vec{A} \cdot \vec{B} = AB \cos \theta$ with $|\vec{B}| = \sqrt{1 + 1 + 16} = \sqrt{18}$:

$$\cos \theta = \frac{3}{\sqrt{14}\sqrt{18}} \approx 0.189 \implies \theta \approx 79^\circ.$$

1.5 Position, Displacement, and Separation Vectors

Position vector

A point in space is described by its Cartesian coordinates: (x, y) in 2-D or (x, y, z) in 3-D. The vector from the origin O to that point is the **position vector**:

$$\vec{r} \equiv x\hat{x} + y\hat{y} + z\hat{z}.$$

Its magnitude is the distance from the origin, $r = \sqrt{x^2 + y^2 + z^2}$, and the unit vector pointing radially outward is

$$\hat{r} = \frac{\vec{r}}{r} = \frac{x\hat{x} + y\hat{y} + z\hat{z}}{\sqrt{x^2 + y^2 + z^2}}.$$

Separation vector

Why bother with one more definition? Because next term, in electricity, the force a charge feels depends on where it is *relative to another charge* — not relative to the origin. The vector that does that bookkeeping is worth naming now, while the geometry is fresh.

Separation vector $\vec{\Delta}$

The separation vector is the displacement from a source point $\vec{r}'$ (where something is located, e.g. an electric charge) to a field point $\vec{r}$ (where you are observing):

$$\vec{\Delta} \equiv \vec{r} - \vec{r}'.$$

Its magnitude is $\Delta = |\vec{r} - \vec{r}'|$, and the unit vector pointing from the source to the field point is

$$\hat{\Delta} = \frac{\vec{\Delta}}{\Delta} = \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|}.$$

In Cartesian coordinates,

$$\begin{aligned}\vec{\Delta} &= (x - x')\hat{x} + (y - y')\hat{y} + (z - z')\hat{z}, \\ \Delta &= \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}, \\ \hat{\Delta} &= \frac{(x - x')\hat{x} + (y - y')\hat{y} + (z - z')\hat{z}}{\sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}}.\end{aligned}$$

Worked Example 1.8: A separation vector

A charge sits at the source point $\vec{r}' = (1, 0, 2)$ and we observe at the field point $\vec{r} = (4, 4, 2)$ (coordinates in metres). Then

$$\begin{aligned}\vec{\Delta} &= (4 - 1)\hat{x} + (4 - 0)\hat{y} + (2 - 2)\hat{z} = 3\hat{x} + 4\hat{y}, \\ \Delta &= \sqrt{3^2 + 4^2 + 0^2} = 5 \text{ m}, \quad \hat{\Delta} = \frac{3}{5}\hat{x} + \frac{4}{5}\hat{y}.\end{aligned}$$

When are two vectors equal?

Two vectors are equal only if their corresponding components are equal — equivalently, if they have the same magnitude and the same direction. Sliding a vector to a new position (a parallel translation) does not change it.

Formula Summary: Vectors

Formula Summary: Vectors

Quantity	Formula
Resultant (parallelogram law)	$R = \sqrt{a^2 + 2ab \cos \theta + b^2}$
Direction of resultant	$\alpha = \tan^{-1}\left(\frac{b \sin \theta}{a + b \cos \theta}\right)$
Dot product	$\vec{A} \cdot \vec{B} = AB \cos \theta = A_x B_x + A_y B_y + A_z B_z$
Cross-product magnitude	$ \vec{A} \times \vec{B} = AB \sin \theta$
Components (2-D)	$A_x = A \cos \theta, \quad A_y = A \sin \theta$
Magnitude (2-D)	$A = \sqrt{A_x^2 + A_y^2}$
Magnitude (3-D)	$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$
Unit vector	$\hat{A} = \vec{A}/ \vec{A} $
Position vector	$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$
Separation vector	$\vec{\Delta} = \vec{r} - \vec{r}'$

Dimensional Analysis

Here is a challenge that sounds impossible. Take a pendulum — a stone on a string. Its period (the time for one full swing) surely depends on things like the length of the string, the mass of the stone, how hard gravity pulls. Now the challenge: *predict the formula for the period without using any law of motion at all*. No forces, no Newton, no experiment. Just bookkeeping.

By the end of this short chapter you will do exactly that — and you will discover along the way that the mass of the stone *cannot appear in the answer*, a fact you can then walk outside and test with two stones and a shoelace. The trick is called **dimensional analysis**, and physicists reach for it constantly: to catch algebra mistakes, to guess formulas before deriving them, and sometimes (as the supplementary box at the end of the chapter shows) to uncover military secrets from a published photograph.

Stop and Predict 2.1: The pendulum, before any theory

A pendulum of length l carries a bob of mass m , swinging under gravity g . Before reading on, write down your instincts:

1. If you double the mass of the bob, does the period get longer, shorter, or stay the same?
2. If you double the length of the string, what happens to the period — doubles? quadruples? something else?

Keep your answers. The chapter will grade them — twice: once by bookkeeping, once (if your class has a shoelace) by experiment.

2.1 Dimension

In physics, we measure many different things, but they are all built from a small set of seven fundamental base quantities (see Table 2.1).

Table 2.1: The seven fundamental base quantities, their dimensions, and sample units.

Base Quantity	Dimension	Sample Units
Mass	$[M]$	kg, g, lbs
Length	$[L]$	m, cm, feet, miles
Time	$[T]$	s, minutes, hours, years
Electric Current	$[I]$	A, mA
Thermodynamic Temperature	$[\Theta]$	K, °C, °F
Amount of Substance	$[N]$	mol
Luminous Intensity	$[J]$	cd

The **dimension** of any physical quantity represents the powers to which these base quantities must be raised to express that quantity. Example:

$$[\text{Force}] = [\text{mass}] \times \frac{[\text{velocity}]}{[\text{time}]} = [\text{mass}] \times \frac{[\text{length}/\text{time}]}{[\text{time}]} = MLT^{-2}.$$

Such an expression for a physical quantity in terms of the base quantities is called the **dimensional formula**. Thus, the dimensional formula of force is MLT^{-2} . In other words, we say the dimensions of force are 1 in mass, 1 in length, and -2 in time. The dimensions in all other base quantities are zero.

Note that in this type of calculation the magnitudes are not considered. It is equality of the *type* of quantity that enters. Thus, change in velocity, initial velocity, average velocity, and final velocity are all equivalent in this discussion — each one is LT^{-1} .

The physical quantity that is expressed in terms of the base quantities is enclosed in square brackets to remind us that the equation is among the dimensions and not among the magnitudes.

2.2 The Principle of Homogeneity

You cannot add apples to oranges, and in physics, you cannot add velocity to force.

The Principle of Homogeneity

The dimensions of all separated terms in a physical equation must be identical.

Let us check the kinematic equation $x = ut + \frac{1}{2}at^2$:

- $[x] = L$
- $[ut] = (LT^{-1})(T) = L$
- $[\frac{1}{2}at^2] = (LT^{-2})(T^2) = L$

Since every term evaluates to length $[L]$, the equation is dimensionally correct. Note: the constant $\frac{1}{2}$ has no dimension and is ignored in this check.

Checkpoint 2.1: Spot the impostor

A student solving a problem about a falling stone produces four candidate formulas, where v is a speed, g an acceleration, h a height, and t a time. Exactly one of them *could* be right. Which one?

- (a) $v = 2gh$
- (b) $v = \sqrt{2gh}$
- (c) $h = \frac{1}{2}gt$
- (d) $t = \sqrt{2hg}$

Answer: (b). Check each: $[gh] = (LT^{-2})(L) = L^2T^{-2}$, which is speed squared — so (a) fails but its square root (b) has dimension LT^{-1} , a speed. In (c), $[gt] = LT^{-1}$ is a speed, not a height. In (d), $[v] = \sqrt{L^2T^{-2}} = LT^{-1}$, again a speed, not a time. Note what the check cannot do: it cannot tell you whether the 2 in (b) is right.

2.3 Uses of Dimensional Analysis

Checking the Correctness of an Equation

As shown above, if an equation is not dimensionally homogeneous, it is physically impossible and strictly wrong.

Limitation: A dimensionally correct equation might still be wrong if numerical constants are missing.

Conversion of Units

Dimensions help us find conversion factors between different unit systems (like SI and CGS). For example, to convert 1 Pascal (SI unit of pressure) to CGS pressure:

$$[\text{Pressure}] = \frac{[\text{Force}]}{[\text{Area}]} = \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}.$$

$$1 \text{ Pascal} = (1 \text{ kg})(1 \text{ m})^{-1}(1 \text{ s})^{-2}, \quad 1 \text{ CGS pressure} = (1 \text{ g})(1 \text{ cm})^{-1}(1 \text{ s})^{-2}.$$

$$\frac{1 \text{ Pascal}}{1 \text{ CGS pressure}} = \left(\frac{1 \text{ kg}}{1 \text{ g}}\right) \left(\frac{1 \text{ m}}{1 \text{ cm}}\right)^{-1} \left(\frac{1 \text{ s}}{1 \text{ s}}\right)^{-2} = (10^3)(10^2)^{-1} = 10.$$

So 1 Pascal = 10 CGS pressure.

Deducing Relations Among Physical Quantities

This is the payoff promised at the start of the chapter, and it has a history. Lord Rayleigh — the physicist who explained why the sky is blue — used this “method of dimensions” so often and so effectively in the late 1800s that it became a standard weapon in every physicist’s toolkit; he was still writing notes to *Nature* in 1915 urging that the method could save “much labour” by anticipating results before any hard analysis. The idea: if we know *which* variables affect a physical quantity, the requirement that the dimensions balance often pins down *how* they can combine, up to a dimensionless constant.

Time to grade your pendulum predictions. Suppose the time period t of a simple pendulum depends on its length l , the mass of the bob m , and the acceleration due to gravity g . We assume a product relation of the form

$$t = k l^a m^b g^c, \quad (1)$$

where k is a dimensionless constant and a, b, c are exponents we want to evaluate. Taking dimensions on both sides:

$$T = L^a M^b (LT^{-2})^c = L^{a+c} M^b T^{-2c}.$$

Equating powers on both sides:

- For M : $b = 0$
- For T : $-2c = 1 \implies c = -\frac{1}{2}$
- For L : $a + c = 0 \implies a = \frac{1}{2}$

Substituting these values gives

$$t = k \sqrt{\frac{l}{g}}. \quad (2)$$

In the actual formula $k = 2\pi$, which is determined from the full physical analysis, not from dimensional analysis alone.

Look at what the bookkeeping just told us. The mass exponent came out $b = 0$: *the mass of the bob cannot appear in the formula*. A heavy stone and a light stone on equal strings must swing in step — not because we measured it, but because no other answer can balance dimensionally. (If your first instinct in Stop and Predict 2.1 was “heavier swings slower,” you are in good company — most people guess that. Now tie two different stones to equal shoelaces and watch.) And doubling the length does not double the period; it multiplies it by $\sqrt{2} \approx 1.4$, because the length enters under a square root. This “predict by bookkeeping, then test” cycle is exactly how you will verify the formula in the lab later this year.

Limitations:

1. It only works for *product-type* dependencies. It cannot deduce equations where terms are added or subtracted (like $x = ut + \frac{1}{2}at^2$).
2. It cannot find numerical constants. The dimensional method left us with an unknown constant k in the pendulum equation. We need actual experiments (or deeper theory) to find that $k = 2\pi$.

Supplementary — How dimensional analysis leaked an atomic secret
(optional — read for pleasure, skip without loss)

In 1945 the United States exploded the first atomic bomb in the New Mexico desert. The energy released — the “yield” — was one of the most closely guarded secrets in the world. Two years later, the government declassified something that seemed

harmless: a filmstrip of photographs of the growing fireball, each frame stamped with the time after detonation and a distance scale.

The British physicist G. I. Taylor looked at those photographs and reasoned like this. Once the blast is underway, the radius R of the fireball can plausibly depend on only three things: the energy E released, the density ρ of the surrounding air, and the time t since detonation. There is exactly one way to combine $[E] = ML^2T^{-2}$, $[\rho] = ML^{-3}$, and $[T]$ into a length (try it — it is the same game we played with the pendulum):

$$R \sim \left(\frac{E t^2}{\rho} \right)^{1/5} \quad \Longleftrightarrow \quad E \sim \frac{\rho R^5}{t^2}.$$

Taylor read R and t off the published frames (about 140 m at $t = 25$ ms), put in $\rho \approx 1.2$ kg/m³ for air, and out came $E \sim 10^{14}$ J — the equivalent of roughly twenty thousand tonnes of TNT. When he published the estimate in 1950 it was embarrassingly close to the still-classified value. The American military was reportedly not amused; no law had been broken, only bookkeeping applied. *Wonder en is gheen wonder.*

Motion

Twenty-four centuries ago the Greek philosopher Zeno fired an arrow — in an argument — and claimed it could not possibly be moving. His reasoning: freeze the arrow at any single instant, and it occupies exactly one place, just like an arrow at rest. If it is indistinguishable from a resting arrow at *every* instant, when exactly does it do its moving?

You know the arrow moves. But Zeno's question is sharper than it looks, and it survived for two thousand years because answering it properly requires an idea nobody had: a precise meaning for *how fast something is moving at a single instant*. Speed is distance divided by time — but at one instant, no distance is covered and no time passes. 0/0. What then does the number on a car's speedometer actually *mean*?

This chapter builds that missing idea. It is called the derivative, it was invented for exactly this problem (by Newton and Leibniz, independently, in the 1660s–70s), and it is the tool that finally lets us answer Zeno: motion is not a property of one frozen instant, but of how neighbouring instants compare — and there is a precise way to take that comparison to the limit.

3.1 What is Motion?

Motion and Rest

The apparent change in a body's (or object's) position with time is called **motion**. If the body does not change its position as time passes, it is said to be at **rest**.

The relative nature of motion

But when do we say that a body is not changing its position? A cup on the table remains on the table — we say the cup is at rest. However, if we station ourselves on Mars, the whole Earth is found to be changing its position, and so the room, the table, and the cup are all continuously changing their positions.

- The cup is *at rest* if viewed from the room.
- The cup is *moving* if viewed from Mars.

Motion is therefore a combined property of the object under study and the observer. There is no meaning of rest or motion without the viewer.

Key Principle

Nothing is in absolute rest or in absolute motion.
Mars is moving with respect to the cup; the cup moves with respect to Mars.

Worked Example 3.1: Frames and relative motion

A passenger sits still inside a moving train.

- (a) From the passenger's own frame: the passenger is *at rest*.
- (b) From a bystander on the platform: the passenger is *moving* at the speed of the train.
- (c) From an observer on the Moon: both the passenger and the bystander are moving (the Earth rotates and orbits the Sun).

All three descriptions are equally valid — rest and motion depend entirely on the reference frame of the observer.

Frame of reference

To locate the position of a particle we need a **frame of reference** — a choice of a spatial origin and coordinate axes to label position. We choose Cartesian coordinates, so the coordinates (x, y, z) of the particle P specify its position relative to that frame. A clock is added to the frame to measure time.

Rest and motion in terms of coordinates

Rest: If all three coordinates x, y, z of the particle remain unchanged as time passes, the particle is at rest with respect to this frame.

Motion: If any one or more coordinates change with time, the body is moving with respect to this frame.

One-dimensional motion first

For our first look at this subject we will forget about the three dimensions of the world and concentrate on moving in one direction — like a car on a single straight road. We will return to three dimensions after we see how to describe motion in one dimension.

3.2 Velocities and the Derivative

Position function

Suppose an object moves along a straight line. We describe its motion by an **equation of motion**

$$s = f(t),$$

where s is the displacement (direct distance from the origin along the line) of the object at time t . The function f that describes the motion is called the **position function** of the object.

Average velocity

In the time interval from t to $t + \Delta t$, the change in position (displacement) is $f(t + \Delta t) - f(t)$.

Average velocity

The average velocity over the time interval $[t, t + \Delta t]$ is

$$v_{\text{av}} = \frac{\text{displacement}}{\text{time}} = \frac{f(t + \Delta t) - f(t)}{\Delta t}.$$

Geometrically, this is the slope of the secant line PQ on the position-time graph.

Worked Example 3.2: Computing average velocity

A car moves along a straight road with position function $f(t) = 3t^2 + 2t$ metres (where t is in seconds).

(a) Displacement from $t = 1$ s to $t = 3$ s:

$$f(1) = 3(1)^2 + 2(1) = 5 \text{ m}, \quad f(3) = 3(3)^2 + 2(3) = 33 \text{ m}, \quad \Delta s = 28 \text{ m}.$$

(b) Average velocity over $[1, 3]$:

$$v_{\text{av}} = \frac{f(3) - f(1)}{3 - 1} = \frac{28}{2} = 14 \text{ m/s}.$$

Try It Yourself 3.1: Average velocity from a table

A cyclist's position at various times is recorded below. Find the average velocity over the full 15 s interval.

t (s)	0	5	10	15
s (m)	0	12	20	24

(Answer: $v_{\text{av}} = 24/15 = 1.6 \text{ m/s}$.)

Instantaneous velocity

Average velocity is honest but blunt. A traffic officer stops a driver: “You were doing ninety kilometres per hour.” The driver protests: “That’s impossible — I’ve only been driving for ten minutes, nowhere near an hour!” The driver is joking (we hope), but hiding in the joke is Zeno’s question from the start of the chapter: “90 km/h *at this moment*” cannot mean that any hour, or any minute, was actually timed. It must mean something like: *if* the car kept doing what it is doing right now, it *would* cover 90 km in an hour. To make that precise, we shrink the stopwatch window.

So we compute average velocities over shorter and shorter time intervals $[t, t + \Delta t]$ — in other words, we let Δt approach zero. The averages settle toward a definite number,

and *that number* is what the speedometer reports.

Instantaneous velocity

The instantaneous velocity $v(t)$ at time t is defined as the limit of these average velocities:

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t} = \frac{df}{dt},$$

provided that this limit exists. Geometrically, $v(t)$ equals the slope of the tangent line ℓ to the position-time curve at the point P .

Worked Example 3.3: Instantaneous velocity from the limit definition

Find the instantaneous velocity $v(t)$ of the car $f(t) = 3t^2 + 2t$ from the limit definition. Hence find $v(1)$ and $v(3)$.

Solution. Expand $f(t + \Delta t)$:

$$f(t + \Delta t) = 3(t + \Delta t)^2 + 2(t + \Delta t) = 3t^2 + 6t \Delta t + 3(\Delta t)^2 + 2t + 2 \Delta t.$$

Subtract $f(t) = 3t^2 + 2t$:

$$f(t + \Delta t) - f(t) = 6t \Delta t + 3(\Delta t)^2 + 2 \Delta t = \Delta t (6t + 3 \Delta t + 2).$$

Divide by Δt and take $\Delta t \rightarrow 0$:

$$v(t) = \lim_{\Delta t \rightarrow 0} (6t + 3 \Delta t + 2) = 6t + 2 \text{ m/s}.$$

At $t = 1$ s: $v(1) = 6(1) + 2 = 8$ m/s. At $t = 3$ s: $v(3) = 6(3) + 2 = 20$ m/s.

Note: the average velocity over $[1, 3]$ was 14 m/s, which lies between 8 and 20 m/s — exactly as expected!

Try It Yourself 3.2: Limit definition for $f(t) = t^2$

Use the limit definition to show that for $f(t) = t^2$, the instantaneous velocity is $v(t) = 2t$ m/s. Hence find the velocity at $t = 3$ s.

Hint: $(t + \Delta t)^2 - t^2 = 2t \Delta t + (\Delta t)^2$. *Answer:* $v(3) = 6$ m/s.

The derivative

The quantity df/dt found above is called the **derivative** of f with respect to t . The process of finding it is called **differentiating**.

Tangent line

The tangent line to $y = f(t)$ at the point $(t, f(t))$ is the unique line through $(t, f(t))$ whose slope equals $\frac{df(t)}{dt}$, the derivative of f at t .

Other notations for the derivative

Let $y = f(x)$. All of the following mean the same thing:

$$\frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = f'(x) = y'.$$

For derivatives with respect to time t , Newton used an overdot:

$$\dot{f} = \frac{df}{dt}, \quad \dot{y} = \frac{dy}{dt}.$$

Notation	Name / origin
$\frac{dy}{dx}, \frac{df}{dx}, \frac{d}{dx}f(x)$	Leibniz notation
$f'(x), y'$	Lagrange (prime) notation
$\dot{y}, \dot{f}$	Newton (dot) notation — time derivatives

All notations are in common use. In physics, $\dot{s} = v$ for velocity is popular for time derivatives. You should be comfortable reading all of them.

Formula Summary: Motion

Formula Summary: Motion	
Quantity	Formula
Position function	$s = f(t)$
Displacement over $[t, t + \Delta t]$	$\Delta s = f(t + \Delta t) - f(t)$
Average velocity	$v_{\text{av}} = \frac{f(t + \Delta t) - f(t)}{\Delta t}$
Instantaneous velocity	$v(t) = \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t} = \frac{df}{dt}$
Slope of tangent	$\frac{df}{dt}$ at that point

An important note on vectors: Because we are currently studying motion along a single straight line (1D), the direction of our vectors is indicated simply by a positive or negative sign. To keep the focus on the calculus, we will temporarily drop the vector notation ($\vec{v}$) and use scalars (v) to represent these 1D quantities.

3.3 Velocity as a Derivative and Graphing

Ex 1. A falling ball ($s = f(t) = 5t^2$).

t (s)	s (m)
0	0
1	5
2	20
3	45
4	80
5	125

The formula for this curve can be written as $s = f(t) = 5t^2$, which enables us to calculate the distance at any time.

If we want to know how fast a ball is falling at exactly $t = 4$ s, we cannot simply use a wide interval. Between 3 and 4 seconds, the ball averages 35 m/s ($v_{\text{av}} = \frac{80-45}{4-3}$). However, it is constantly accelerating during that time. That average does not reflect the true velocity at the exact 4-second mark.

To get closer to the truth, we look at a very small interval immediately following $t = 4$ s:

- At 4.1 s: total displacement $s = 5(4.1)^2 = 84.05$ m.
- At 4 s: total displacement was 80 m.
- Average velocity for this tiny window: $4.05/0.1 = 40.5$ m/s.

Taking the general approach, if at time $t_0 = 4$ s the position is $s_0 = 5t_0^2$, then at $t_0 + \Delta t$ the new position is

$$s_0 + \Delta s = 5(t_0 + \Delta t)^2 = 5t_0^2 + 10t_0 \Delta t + 5 \Delta t^2.$$

So the extra displacement is $\Delta s = 10t_0 \Delta t + 5 \Delta t^2$, and the average velocity is

$$v_{\text{av}} = \frac{\Delta s}{\Delta t} = 10t_0 + 5 \Delta t. \quad (3)$$

Taking $\Delta t \rightarrow 0$:

$$v(\text{at time } t_0) = 10t_0.$$

At $t_0 = 4$ s: $v(4) = 10 \times 4 = 40$ m/s.

The procedure we have just carried out is performed so often in mathematics that special notation has been assigned. The velocity is

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}. \quad (4)$$

The quantity ds/dt is called the “derivative of s with respect to t ”. For our example:

$$\frac{d}{dt}(5t^2) = 10t.$$

A Short Table of Derivatives.

u, v, w are arbitrary functions of t ; c and n are arbitrary constants.

Function, $f(t)$	Derivative, $\frac{df}{dt}$
c (constant)	0
t^n	$n t^{n-1}$
cu	$c \frac{du}{dt}$
$\sin(t)$	$\cos(t)$
$\cos(t)$	$-\sin(t)$
e^t	e^t
$\ln(t)$	$\frac{1}{t}$
$u + v + w + \cdots$	$\frac{du}{dt} + \frac{dv}{dt} + \frac{dw}{dt} + \cdots$
uv	$u \frac{dv}{dt} + v \frac{du}{dt}$
$\frac{u}{v}$	$\frac{v (du/dt) - u (dv/dt)}{v^2}$

Exercise. Find the derivative of the following functions.

- a. $f(t) = 3t - 8$
- b. $f(x) = mx + b$
- c. $f(t) = 2.5t^2 + 6t$
- d. $f(x) = 4 + 8x - 5x^2$
- e. $A(p) = 4p^3 + \ln(p)$
- f. $F(t) = e^t(t^2 - 1)$
- g. $f(x) = \frac{1}{x^2 - 4}$
- h. $F(v) = \frac{v}{v + 2}$
- i. $g(u) = \frac{u + 1}{4u - 1}$
- j. $f(x) = x^4 + \sin(x)$
- k. $g(x) = \frac{1}{1 + \sqrt{x}}$

Example solutions:

$$(a) \quad f'(t) = 3, \quad (d) \quad f'(x) = 8 - 10x.$$

3.4 Distance as an Integral

Now we discuss the inverse problem. Suppose that instead of a table of distances, we have a table of speeds at different times, starting from zero. From our earlier result, $s = 5t^2$ and $v = 10t$.

The distance travelled is equal to the area under the velocity–time curve. In general, when the velocity–time graph is not a simple geometric shape, we break the motion into small intervals Δt and approximate the distance in each interval as $\Delta s \approx v \Delta t$.

In general, suppose an object moves with velocity $v = v(t)$, where $a \leq t \leq b$. We take velocity readings at times $t_0(=a), t_1, t_2, \dots, t_n(=b)$ so that the velocity is approximately constant on each sub-interval with $\Delta t = (b-a)/n$. The total distance is approximately

$$v(t_0)\Delta t + v(t_1)\Delta t + \dots + v(t_{n-1})\Delta t = \sum_{i=0}^{n-1} v(t_i) \Delta t.$$

The true distance is the limit as $\Delta t \rightarrow 0$ (equivalently, $n \rightarrow \infty$):

$$s = \lim_{\Delta t \rightarrow 0} \sum_{i=0}^{n-1} v(t_i) \Delta t = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} v(t_i) \Delta t. \quad (5)$$

The mathematicians have invented a symbol for this limit. The Δ turns into a d , the addition is written with a great “S” (from the Latin *summa*), distorted into the integral sign:

$$s = \int v(t) dt. \quad (6)$$

This process of adding all these terms together is called **integration**, and it is the opposite process to differentiation.

A Short Table of Derivatives and Integrals.

<i>u, v, w are arbitrary functions of t; c, k, n are arbitrary constants.</i>		
Function, $f(t)$	Derivative, $\frac{df}{dt}$	Integral, $\int f(t) dt$
c	0	$ct + k$
t^n	$n t^{n-1}$	$\frac{t^{n+1}}{n+1} + k \quad (n \neq -1)$
cu	$c \frac{du}{dt}$	$c \int u dt$
$\sin(t)$	$\cos(t)$	$-\cos(t) + k$
$\cos(t)$	$-\sin(t)$	$\sin(t) + k$
e^t	e^t	$e^t + k$
$u + v + w + \dots$	$\frac{du}{dt} + \frac{dv}{dt} + \dots$	$\int u dt + \int v dt + \dots$

Returning to our falling ball example, $v(t) = 10t$:

$$s = \int v(t) dt = \int 10t dt = 5t^2 + C.$$

The distance travelled between $t = 0$ and $t = 5$ s:

$$s = \int_0^5 v(t) dt = \int_0^5 10t dt = [5t^2]_0^5 = 5(5)^2 - 5(0)^2 = 125 \text{ m},$$

which agrees with the area under the velocity–time graph over $[0, 5]$.

Exercise. Evaluate the integral.

a. $\int (3t - 8) dt$

b. $\int (mx + b) dx$

c. $\int_0^3 (5t^2 + 6t) dt$

d. $\int_{\pi}^4 (4 + 8x - 5x^2) dx$

e. $\int_1^2 (3e^t + (t^2 - 1)) dt$

f. $\int_0^{\pi} (x^4 + \sin x) dx$

Example solutions:

$$(a) \int (3t-8) dt = \frac{3}{2}t^2 - 8t + k, \quad (c) \int_0^3 (5t^2+6t) dt = \frac{5}{3}(3^3-0^3) + 3(3^2-0^2) = 45+27 = 72.$$

3.5 Acceleration

If the velocity of a particle remains constant as time passes, we say that it is moving with **uniform velocity**. If the velocity changes with time, the particle is said to be **accelerating**.

The **average acceleration** over $[t_1, t_2]$ is

$$a_{\text{av}} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t}. \quad (7)$$

The **instantaneous acceleration** is

$$a(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}. \quad (8)$$

Since velocity is ds/dt , and acceleration is the time derivative of velocity:

$$a(t) = \frac{dv}{dt} = \frac{d}{dt} \left(\frac{ds}{dt} \right) = \frac{d^2s}{dt^2}, \quad (9)$$

which is the **second derivative** of position with respect to time.

For the falling ball $s(t) = 5t^2$:

$$v(t) = \frac{ds}{dt} = 10t, \quad a(t) = \frac{dv}{dt} = 10 \text{ m/s}^2.$$

The acceleration is constant because the gravitational force on the falling body is constant, and by Newton's second law, acceleration is proportional to the net force.

On the companion site: the **motion grapher** (Chapter 3 on the course website) draws $s(t)$, $v(t)$, and $a(t)$ one above the other for motions you choose, with a dot moving along the track. Before revealing each graph, sketch your guess — slope-reading is a skill, and it is trained by being wrong in private.

Checkpoint 3.1: The top of the flight

You toss a ball straight up. At the very highest point of its flight, its velocity is zero. What is its acceleration there?

- (a) Zero — the ball is momentarily at rest.
- (b) 9.8 m/s^2 downward, same as everywhere else in the flight.
- (c) 9.8 m/s^2 upward, since the ball is still rising until that moment.
- (d) Cannot be determined without knowing the ball's mass.

Answer: (b). Velocity and acceleration are different questions: "how fast is it moving?" versus "how fast is its velocity changing?" At the top, $v = 0$ for one instant, but v is changing at the same steady rate as always — from small-upward, through zero, to small-downward. If the acceleration were really zero there, the velocity would stop changing, and the ball would hang in the air forever.

3.6 Motion with Constant Acceleration

Suppose a particle moves along a straight line (taken as the x -axis) with constant acceleration a . Let the velocity at $t = 0$ be v_0 and at time t be v . Then:

$$\frac{dv}{dt} = a \implies dv = a dt.$$

Integrating from 0 to t (velocity from v_0 to v):

$$\int_{v_0}^v dv = \int_0^t a dt \implies v - v_0 = at, \quad \boxed{v = v_0 + at.} \quad (10)$$

Writing $dx/dt = v_0 + at$ and integrating from x_0 to x (time from 0 to t):

$$x - x_0 = v_0 t + \frac{1}{2} at^2,$$

$$\boxed{\Delta x = v_0 t + \frac{1}{2} a t^2.} \quad (11)$$

Squaring equation (10): $v^2 = (v_0 + at)^2 = v_0^2 + 2av_0t + a^2t^2 = v_0^2 + 2a(v_0t + \frac{1}{2}at^2)$, so

$$\boxed{v^2 = v_0^2 + 2a \Delta x.} \quad (12)$$

Equations of Kinematics

$$v = v_0 + at, \quad \Delta x = v_0 t + \frac{1}{2} a t^2, \quad v^2 = v_0^2 + 2a \Delta x.$$

The three equations above are called the **kinematic equations**. Remember that x represents the position (or displacement) at time t and not necessarily the total distance travelled. The quantities v_0 , v , and a may take positive or negative values depending on their direction.

3.7 Freely Falling Bodies

A common example of motion in a straight line with constant acceleration is the **free fall** of a body near the earth's surface. If air resistance is neglected and a body is dropped, it falls along a vertical straight line. The acceleration in the vertically downward direction has magnitude approximately 9.8 m/s^2 and is known as the **acceleration due to gravity**, denoted by g .

The sign of g depends on the chosen positive direction. Taking vertically upward as positive gives $a = -g$; taking downward as positive gives $a = +g$. Substituting into the kinematic equations accordingly:

$$v = v_0 \pm g t, \quad \Delta y = v_0 t \pm \frac{1}{2} g t^2, \quad v^2 = v_0^2 \pm 2g \Delta y,$$

where the upper sign (+) applies when downward is taken as positive, and the lower sign (−) when upward is taken as positive.

Motion in a Plane (2D) and in Space (3D). In three dimensions the position vector is

$$\vec{s} = x\hat{x} + y\hat{y} + z\hat{z}.$$

Differentiating with respect to time gives the velocity vector:

$$\vec{v} = \frac{d\vec{s}}{dt} = \frac{dx}{dt}\hat{x} + \frac{dy}{dt}\hat{y} + \frac{dz}{dt}\hat{z} = v_x\hat{x} + v_y\hat{y} + v_z\hat{z},$$

with magnitude (speed)

$$v = |\vec{v}| = \sqrt{v_x^2 + v_y^2 + v_z^2}.$$

Differentiating once more gives the acceleration vector:

$$\vec{a} = \frac{d\vec{v}}{dt} = a_x\hat{x} + a_y\hat{y} + a_z\hat{z}, \quad a = \sqrt{a_x^2 + a_y^2 + a_z^2}.$$

3.8 Projectile Motion

An important example of motion in a plane with constant acceleration is **projectile motion**. When a particle is thrown obliquely near the earth's surface, it moves along a curved path. Such a particle is called a **projectile**. We assume the particle remains close to the surface and air resistance is negligible, so acceleration is constant in the vertically downward direction with magnitude $g \approx 9.8 \text{ m/s}^2$.

Stop and Predict 3.1: Two coins

Put two identical coins at the edge of a table. At the same moment, let one simply *drop* off the edge, and *flick* the other horizontally so it flies off with some sideways speed. Which coin hits the floor first? Commit to a prediction — then actually do it (listen for one click or two).

The result surprises almost everyone the first time: they land *together*. The flicked coin travels metres sideways, yet loses height at exactly the same rate as the dropped one. Gravity does not care about horizontal motion. That single observation is the key to everything in this section: projectile motion is just two *independent* motions happening at once — constant velocity horizontally, free fall vertically — and we may solve each with the one-dimensional tools we already own.

A particle is projected from point O with initial speed v_0 at angle θ with the horizontal. The initial conditions are:

$$x_0 = 0, \quad v_{0,x} = v_0 \cos \theta, \quad a_x = 0, \quad (13)$$

$$y_0 = 0, \quad v_{0,y} = v_0 \sin \theta, \quad a_y = -g. \quad (14)$$

Horizontal Motion ($a_x = 0$):

$$v_x = v_0 \cos \theta \quad (\text{constant}), \quad (15)$$

$$x = v_0 t \cos \theta. \quad (16)$$

Vertical Motion:

$$v_y = v_0 \sin \theta - g t, \quad (17)$$

$$y = v_0 t \sin \theta - \frac{1}{2} g t^2, \quad (18)$$

$$v_y^2 = (v_0 \sin \theta)^2 - 2g y. \quad (19)$$

Time of Flight. At point B (landing), $y = 0$:

$$0 = v_0 t \sin \theta - \frac{1}{2} g t^2 = t(v_0 \sin \theta - \frac{1}{2} g t).$$

The non-trivial solution gives the time of flight:

$$T = \frac{2v_0 \sin \theta}{g}. \quad (20)$$

Range. The horizontal range OB is the distance travelled during time T :

$$OB = v_0 T \cos \theta = \frac{2v_0^2 \sin \theta \cos \theta}{g} = \frac{v_0^2 \sin 2\theta}{g}. \quad (21)$$

Notice that the greatest range is achieved if the projectile is thrown at 45° to the horizontal.

Maximum Height Reached. At the peak, $v_y = 0$. From (18): $t = v_0 \sin \theta / g$. Substituting into (19):

$$H = \frac{v_0^2 \sin^2 \theta}{2g}. \quad (22)$$

The time to reach maximum height is exactly half the total time of flight.

Equation of Trajectory. Eliminating t between (16) and (19):

$$y = x \tan \theta - \frac{g x^2}{2v_0^2 \cos^2 \theta}. \quad (23)$$

This is the equation of the trajectory — a **parabola**.

Newton's Three Laws of Motion

Before stating any law, it is worth asking a simple question. What does it take for something to keep moving? If you roll a ball across the floor, what eventually stops it?

The instinctive answer, and the one given by Aristotle for well over a thousand years, is that motion requires a continuous sustaining force. Remove the force and the motion should stop. This view matches what we see every day: carts roll to a halt, thrown balls fall back, skidding cars come to rest. The trouble is that it is wrong. What actually stops a rolling ball is friction, not the absence of some hidden push. In the absence of friction (or any other opposing force), the ball would continue rolling forever. This is the content of Newton's first law, but it took from Aristotle to Galileo to articulate it clearly, and from Galileo to Newton to place it in its proper setting.

How did Galileo see past two thousand years of common sense? Not by removing friction — he could not — but by a thought experiment so clean you can run it in your head right now. Release a ball on one incline and let it roll down and up a facing incline (Figure 4.1). Galileo observed that the ball climbs back to very nearly its starting height, and reasoned that it would reach *exactly* that height if friction could be removed. Now make the second incline shallower: the ball must roll *farther* to regain its height. Shallower still: farther still. Finally, lay the second incline perfectly flat. The ball still “tries” to regain its height — but now it never can. So it must roll on, at constant speed, *forever*. Nothing is pushing it. Motion needs no push to continue; it only needs nothing to stop it.

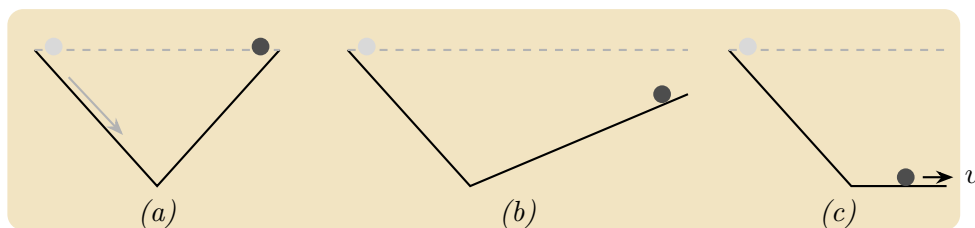


Figure 4.1: Galileo's thought experiment. (a) A ball released on one incline climbs the facing incline back to its starting height. (b) On a shallower incline it rolls farther to regain the same height. (c) On a flat track it can never regain its height — so it rolls forever at constant speed.

4.1 The First Law (the Law of Inertia)

The First Law

In the absence of a net external force, a body at rest remains at rest, and a body in motion continues to move at constant velocity.

The tendency of a body to continue in its initial state of motion is called its **inertia**. Accordingly, the First Law is often called the **Law of Inertia**. Written as an equation:

$$\sum \vec{F} = 0 \iff \vec{a} = 0. \quad (24)$$

This is less a statement about motion than a statement about force: force is not what *maintains* motion, but what *changes* it. A body travelling at constant velocity is no more mysterious than a body at rest — both are states of zero acceleration, and both require zero net force.

There is a subtlety in the first law that is easy to miss. The first law holds only in an **inertial frame** — a frame that is not itself accelerating. The car that brakes suddenly is not such a frame; viewed from the ground, the ball on the dashboard is simply continuing in a straight line at constant velocity while the car decelerates around it.

Is the Earth an inertial frame? Strictly speaking, no. But quantitatively: the Earth's orbital radius is $R \approx 1.5 \times 10^{11}$ m and its orbital period is $T \approx 3.15 \times 10^7$ s, giving an orbital speed

$$v = \frac{2\pi R}{T} \approx 2.99 \times 10^4 \text{ m/s} \quad (25)$$

and a centripetal acceleration

$$a = \frac{v^2}{R} \approx 6 \times 10^{-3} \text{ m/s}^2. \quad (26)$$

Compared to $g = 9.8 \text{ m/s}^2$, this is smaller by a factor of roughly 1600. Newton's first law therefore holds as an extraordinarily good approximation, and we will use it as if it were exact.

Checkpoint 4.1: What pushes the ball?

You toss a ball straight up. While the ball is on its way up, in mid-air, which forces act on it? (Ignore air resistance.)

- (a) Gravity, plus the steadily weakening upward force of your hand.
- (b) Gravity, plus an upward force stored in the ball's motion.
- (c) Gravity only.
- (d) No force at all until the top — then gravity takes over.

Answer: (c). Once the ball leaves your hand, your hand cannot push it — there is no contact. Nothing needs to push the ball upward to keep it moving upward: motion continues by itself (the first law). The one force acting is gravity, which is why the upward motion is steadily eaten away. Aristotle would have picked (a) or (b); most people do, the first time.

4.2 The Second Law

The Second Law

For any particle of mass m , the net force $\vec{F}$ on the particle is always equal to the mass m times the particle's acceleration:

$$\vec{F} = m\vec{a}. \quad (27)$$

Three points are worth emphasising:

1. $\vec{F}$ is the *net* force — the vector sum of *all* forces acting on the body.
2. The equation is a *vector equation*: it must hold component by component, and in particular the acceleration vector always points in the same direction as the net force.
3. Despite appearances, this is *not* a definition of force. Force has an independent physical meaning, and the second law is the (enormously successful) empirical claim that force and acceleration are proportional, with the constant of proportionality being the body's mass.

As a simple application, consider two blocks of mass m_1 and m_2 connected by a light inextensible rope on a frictionless horizontal surface, with a horizontal force F applied to m_2 pulling the pair along. Treating the pair as a single system of mass $M = m_1 + m_2$, the acceleration is

$$a = \frac{F}{m_1 + m_2}. \quad (28)$$

To find the tension in the rope we isolate m_1 . The only horizontal force on m_1 is the tension T , so

$$T = m_1 a = \frac{m_1 F}{m_1 + m_2}. \quad (29)$$

This example illustrates a general strategy. When several bodies are connected, it often pays to treat them first as a single system (to find the common acceleration), and then isolate each body in turn (to find internal forces like tension).

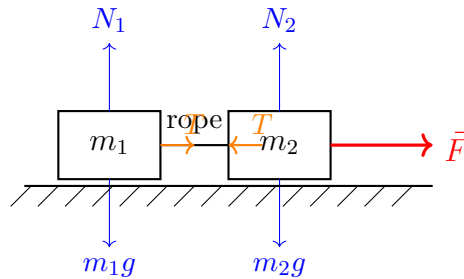


Figure 4.2: Two blocks connected by a light rope on a frictionless surface. Horizontally, m_1 is pulled only by the tension T , while m_2 is pulled forward by F and backward by T .

4.3 The Third Law

The Third Law

If object 1 exerts a force $\vec{F}_{21}$ on object 2, then object 2 will exert a reaction force $\vec{F}_{12}$ on object 1 given by

$$\vec{F}_{12} = -\vec{F}_{21}. \quad (30)$$

The crucial point, and the one most often misunderstood, is that these two forces act on **different bodies**. They therefore never cancel each other.

Let us apply this to a familiar situation. You drop an apple. The Earth pulls the apple downward with gravitational force $F = mg$. By the third law, the apple must pull the Earth upward with an equal force. The apple's acceleration is

$$a_{\text{apple}} = \frac{F}{m_{\text{apple}}} = g \approx 9.8 \text{ m/s}^2. \quad (31)$$

The Earth's acceleration is

$$a_{\text{Earth}} = \frac{F}{M_{\text{Earth}}} = \frac{mg}{M_{\text{Earth}}}. \quad (32)$$

Taking a 100 g apple and $M_{\text{Earth}} \approx 5.97 \times 10^{24} \text{ kg}$:

$$a_{\text{Earth}} \approx 1.6 \times 10^{-25} \text{ m/s}^2. \quad (33)$$

Over the roughly half-second duration of the apple's fall, the Earth moves toward the apple by a distance far smaller than the diameter of a proton.

The principle scales up. The third law is also at the root of the **horse-cart paradox**. A horse pulls a cart forward. By the third law, the cart pulls the horse backward with an equal force. So how does anything ever move? The resolution: the forward pull on the cart and the backward pull on the horse act on *different bodies*. To understand the horse's motion we must look at all forces on the horse, which includes not only the backward pull from the cart but also the forward friction from the ground on its hooves. Both the horse and cart accelerate together. No paradox, only careful bookkeeping.

Checkpoint 4.2: The truck and the mosquito

A speeding truck collides head-on with a mosquito. During the collision, how does the force the truck exerts on the mosquito compare with the force the mosquito exerts on the truck?

- (a) The truck exerts an enormously larger force — just look at the outcome.
- (b) The truck exerts a somewhat larger force, in proportion to its mass.
- (c) The two forces are exactly equal in magnitude.
- (d) The mosquito exerts no force on the truck at all.

Answer: (c). The third law allows no exceptions: the forces in an interaction pair are always equal and opposite. What is wildly unequal is the effect: the same force, by the second law, gives the mosquito an acceleration billions of times larger than the truck's. Equal forces, unequal consequences — do not confuse the force with what the force does.

Class Exercise: The Atwood Machine

In 1784, the Cambridge mathematician George Atwood had a problem: free fall is simply too *fast* to study. A stone dropped from shoulder height is down in half a second — hopeless with an eighteenth-century clock. His solution was beautifully sly: hang two nearly equal masses over a pulley. Gravity still drives the motion, but only the small *difference* between the weights does the driving, while the whole combined mass gets accelerated — so the motion is uniform acceleration in slow motion, gentle enough to time with a pendulum clock. With his machine, Atwood measured g to within about one percent. The apparatus below is the same idea.

Two blocks are connected by a light, inextensible string passing over a small frictionless pulley fixed to the ceiling. Block 1 has mass $m_1 = 6\text{ kg}$ and Block 2 has mass $m_2 = 4\text{ kg}$. The system is released from rest. Take $g = 10\text{ m/s}^2$.

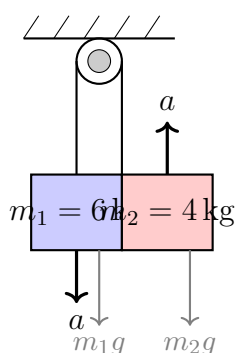


Figure 4.3: The Atwood Machine. Block 1 ($m_1 = 6\text{ kg}$) descends while Block 2 ($m_2 = 4\text{ kg}$) ascends.

Whiteboard Challenge 4.1: The Atwood Machine *(stand up, groups of three, one marker each)*

Before any algebra, as a group, commit to three predictions in one corner of your board:

- (P1) Will the acceleration be bigger than, smaller than, or equal to g ? Why?
- (P2) Will the tension be bigger than, smaller than, or between the two weights $m_1g = 60\text{ N}$ and $m_2g = 40\text{ N}$? Why?
- (P3) If the two masses were *equal*, what would the tension be?

Then earn your answers:

- (a) Draw a free body diagram for each block and write down Newton's second law for each.
- (b) Hence find the acceleration a of the system and the tension T in the string.
- (c) What is the tension in the string if both masses were equal? Comment on your answer physically.

- (d) Check your result against your predictions — and against Atwood’s design goal: show that with $m_1 = 6\text{ kg}$ and $m_2 = 4\text{ kg}$ the acceleration is only $g/5$. Free fall, slowed five-fold.

Further Applications of Newton's Laws

5.1 Contact Forces

Pushing, lifting and pulling are **contact forces** that we experience in the everyday world. Rest your hand on a table; the atoms that form the molecules that make up the table and your hand are in contact with each other. If you press harder, the atoms are also pressed closer together. The electrons in the atoms begin to repel each other and your hand is pushed in the opposite direction by the table.

According to Newton's Third Law, the force of your hand on the table is equal in magnitude and opposite in direction to the force of the table on your hand. Clearly, if you push harder the force increases. If you push your hand straight down on the table, the table pushes back in a direction perpendicular (normal) to the surface. Slide your hand gently forward along the surface of the table. You barely feel the table pushing upward, but you do feel the friction acting as a resistive force to the motion of your hand. This force acts tangential to the surface and opposite to the motion of your hand. Push downward and forward. Try to estimate the magnitude of the force acting on your hand.

The force of the table acting on your hand, $\vec{F}_c \equiv \vec{C}$, is called the **contact force**. This force has both a normal component to the surface, $\vec{C}_\perp \equiv \vec{\mathcal{N}}$, called the **normal force**, and a tangential component to the surface, $\vec{C}_\parallel \equiv \vec{f}$, called the **friction force** (Figure 5.1).

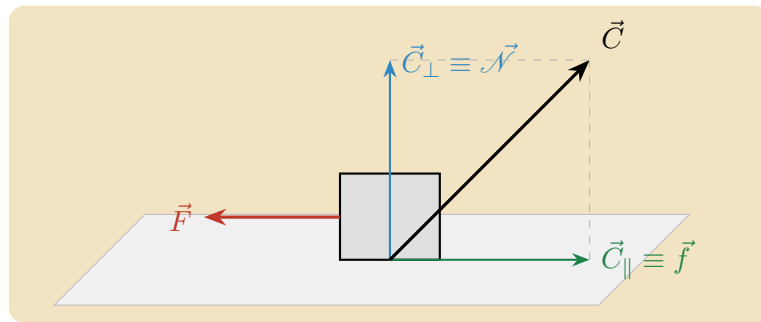


Figure 5.1: The block is pushed to the left by $\vec{F}$. The contact force $\vec{C}$ decomposes into a normal component $\vec{\mathcal{N}}$ and a tangential friction component $\vec{f}$ opposing the motion.

The contact force, written in terms of its component forces, is therefore

$$\vec{C} = \vec{C}_\perp + \vec{C}_\parallel \equiv \vec{\mathcal{N}} + \vec{f}. \quad (34)$$

The normal component $\vec{\mathcal{N}}$ and the tangential component $\vec{f}$ are not independent forces but the vector components of the contact force, perpendicular and parallel to the surface of contact.

Friction

When a block slides along a horizontal surface or down an inclined plane, a lateral force resists the motion. Even when the block is at rest on an inclined plane, a lateral force prevents it from sliding. This resistive force is known as *dry friction*, and there are two distinguishing types when surfaces are in contact with each other.

When two surfaces move relative to one another, the friction is called ***kinetic friction*** (or ***sliding friction***). When the surfaces are stationary relative to each other but a lateral force is present, the friction is called ***static friction***.

Stop and Predict 5.1: The brick, flat or standing?

Take a brick (or this textbook). You can drag it lying *flat* (large face down) or standing on its *edge* (small face down). The weight is the same either way; only the contact area changes — by a factor of three or four. Predict: to slide it at steady speed, which orientation needs the harder pull? Flat, on edge, or no difference? Write your answer down with a one-line reason — then, if a spring scale or a rubber band is handy, test it before reading on.

You have just repeated, with a brick, the experiments Leonardo da Vinci recorded in his notebooks over the years 1493 to about 1515 — the first systematic measurements of kinetic friction ever made. Dragging blocks with hanging weights and pulleys, he found the two facts that still carry his fingerprints today: the force needed to slide a block is proportional to how hard the surfaces are pressed together, and — the surprise — it does *not* depend on the area of contact. Da Vinci never published; his results were rediscovered and published by Guillaume Amontons in 1699, and extended by Charles-Augustin Coulomb, who showed that kinetic friction is also independent of sliding speed for ordinary speeds. Hold on to your brick prediction; the second of da Vinci's facts will grade it in a moment.

Kinetic Friction

The magnitude of the kinetic friction force $\vec{f}^k$ is proportional to the normal force between the two surfaces,

$$f_k = \mu_k \mathcal{N}, \quad (35)$$

where μ_k is the ***coefficient of kinetic friction***, a dimensionless constant depending only on the two materials in contact. Now look at what this law does *not* contain: the contact area. The flat brick and the standing brick need exactly the same pull — if your prediction in Stop and Predict 5.1 said otherwise, you are with the majority, and with common sense; but the experiment (da Vinci's and yours) says no difference. Two blocks of the same mass but different footprints require the same force to slide at constant speed (Figure 5.2). Roughly speaking: the flat brick spreads its weight over more contact points, but presses each one more lightly; the standing brick concentrates the same weight on fewer, harder-pressed points. The two effects cancel. This follows from the microscopic structure of surfaces, as discussed in the next section.

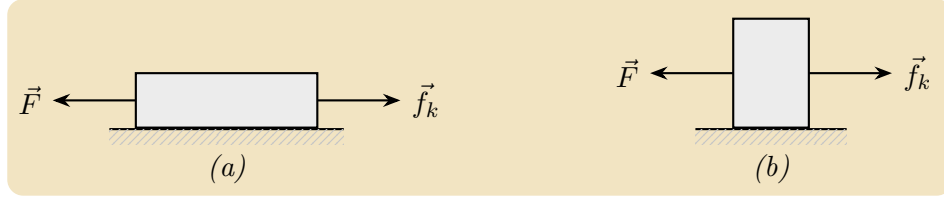


Figure 5.2: Kinetic friction is independent of contact area. Block (a) has twice the footprint of block (b), yet both require the same force to slide at constant speed.

Static Friction

Static friction does not take a fixed value; it adjusts itself to prevent relative motion. As you push a stationary block with increasing force, static friction matches your push exactly until a maximum value is reached. Beyond this threshold, the block slips and kinetic friction takes over. The magnitude of static friction therefore satisfies

$$0 \leq f_s \leq (f_s)_{\max}, \quad (36)$$

where the maximum is found experimentally to be proportional to the normal force,

$$(f_s)_{\max} = \mu_s \mathcal{N}. \quad (37)$$

Here μ_s is the **coefficient of static friction**. Experiment consistently gives $\mu_s > \mu_k$: more force is needed to initiate sliding than to sustain it. This small difference accounts for the stick-slip behaviour of chalk on a blackboard, fingernails on glass, or a violin bow on a string.

The direction of static friction always opposes the applied force, as shown in Figure 5.3.

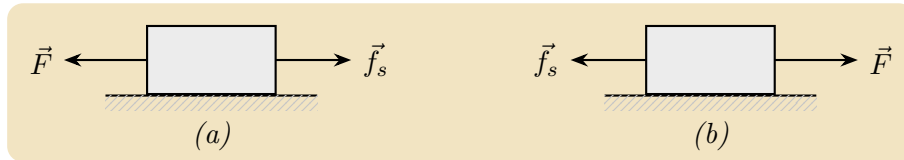


Figure 5.3: Static friction opposes the applied force. In (a) $\vec{F}$ acts to the left so $\vec{f}_s$ acts to the right. In (b) the roles are reversed.

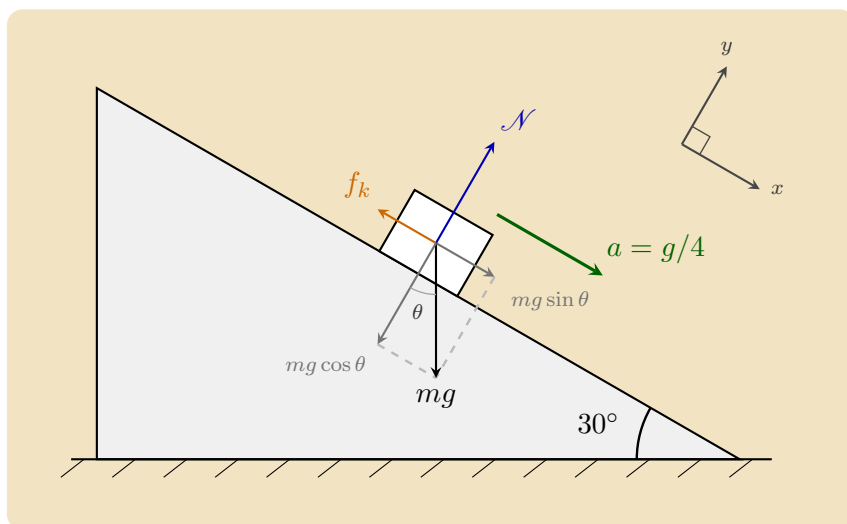
Two important distinctions between static and kinetic friction are worth keeping in mind.

1. The magnitude and direction of static friction depend on the applied forces and adjust to maintain equilibrium, up to the maximum $(f_s)_{\max}$. Kinetic friction, by contrast, has a fixed magnitude $\mu_k \mathcal{N}$ for a given normal force.
2. When the applied force along the contact surface reaches $(f_s)_{\max}$, the surfaces begin to slip. This threshold condition is called the **just slipping** condition, and marks the transition from static to kinetic friction.

Worked Example 5.1: Sliding Block

A block of mass m slides down an incline of angle 30° with an acceleration $g/4$. Find the kinetic friction coefficient μ_k .

Solution. We resolve the forces into components parallel and perpendicular to the incline, taking the positive direction down the slope for motion and outward from the surface for the normal direction.



Perpendicular to the incline there is no acceleration, so the normal force balances the perpendicular component of gravity,

$$\mathcal{N} - mg \cos \theta = 0 \implies \mathcal{N} = mg \cos \theta.$$

Along the incline, the block accelerates down the slope at $a = g/4$. The driving force is the parallel component of gravity, $mg \sin \theta$, opposed by kinetic friction $f_k = \mu_k \mathcal{N}$ acting up the slope. Newton's second law gives

$$mg \sin \theta - \mu_k \mathcal{N} = ma.$$

Substituting $\mathcal{N} = mg \cos \theta$ and $a = g/4$,

$$mg \sin \theta - \mu_k mg \cos \theta = \frac{1}{4}mg.$$

The mass m and the acceleration g cancel throughout, leaving a relation among pure numbers,

$$\sin \theta - \mu_k \cos \theta = \frac{1}{4}.$$

Solving for the coefficient of kinetic friction,

$$\mu_k = \frac{\sin \theta - \frac{1}{4}}{\cos \theta}.$$

With $\theta = 30^\circ$, so that $\sin \theta = \frac{1}{2}$ and $\cos \theta = \frac{\sqrt{3}}{2}$,

$$\mu_k = \frac{1/2 - 1/4}{\sqrt{3}/2} = \frac{1}{2\sqrt{3}}.$$

The Normal Force

When two solid bodies are pressed together, their surfaces resist interpenetration. Push your hand against a wall and the wall pushes back, preventing your hand from passing through (Figure 5.4). This perpendicular push is called the *normal force*, where *normal* means perpendicular to the surface of contact. Notice that the harder you push on the wall with a force $\vec{F}$, the harder the wall pushes back on your hand with the normal force $\vec{\mathcal{N}}$.

The origin of this force is atomic. The atoms at the surface of your hand and the atoms at the surface of the wall repel one another the moment they come into contact, opposing any further compression. The material of the wall behaves, at the microscopic level, like an array of tiny springs.¹ When you push against the wall, these atomic springs compress slightly, and the repulsive force they exert grows with the compression until it exactly balances your push. For a hard material such as concrete, the compression is so small as to be invisible, and the wall appears perfectly rigid.

This is why the normal force is not a fixed quantity: it adjusts itself to match whatever force is pressing the surfaces together. Push harder, and the atomic springs compress more, increasing the normal force in response. As the figure shows, $\vec{F}$ and $\vec{\mathcal{N}}$ always grow together and remain equal in magnitude. The normal force is therefore a *constraint force*, one whose magnitude is determined not by a fixed law but by the condition that the two surfaces do not pass through each other.

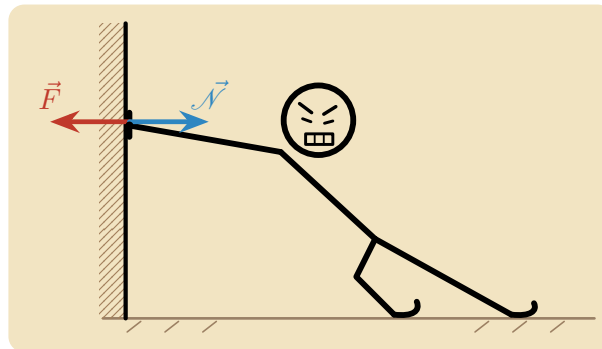


Figure 5.4: When you push against a wall ($\vec{F}$), the wall pushes back on your hand with an equal and opposite normal force ($\vec{\mathcal{N}}$), perpendicular to its surface. Both forces act at the point of contact.

The Normal Force and Weight

We say we are measuring our weight when we step onto a bathroom scale. But what is the scale actually measuring?

Two forces act on a person standing on a scale (Figure 5.1.2). The first is the gravitational force $\vec{F}_g = m\vec{g}$, directed downward. The second is the contact force from the scale;

¹The “atomic springs” picture is a useful analogy but not the complete story. The true origin of this repulsion is quantum mechanical: as electron clouds overlap, the Pauli exclusion principle forces electrons into higher energy states, giving rise to a repulsive force. Classical electrostatics plays a secondary role. We will encounter the Pauli exclusion principle when we study modern physics.

because the person is at rest, this contact force is purely normal and we write it as $\vec{\mathcal{N}}$, directed upward. Since the person is at rest, the vertical acceleration is zero, and Newton's Second Law gives

$$\vec{\mathcal{N}} + \vec{F}_g = \vec{0}. \quad (38)$$

Taking the positive direction upward, this becomes

$$\mathcal{N} - mg = 0 \implies \mathcal{N} = mg. \quad (39)$$

It is worth being precise about four quantities that are easy to confuse. The *true mass* m is the amount of matter in the person, an invariant property that does not change with motion or location.² The *true weight* mg is the actual gravitational force the Earth exerts on the person. The scale, however, does not measure either of these directly. It measures the normal force $\mathcal{N}$ pressing on it, which we call the *apparent weight*. Dividing by g , it displays a number in kilograms, the *apparent mass* $\mathcal{N}/g$. At rest the apparent mass equals the true mass and the distinction seems academic. The elevator, discussed in the next section, will reveal the difference.

The word “weight” is often used loosely to mean the gravitational force the Earth exerts on an object. We shall always call that force the *gravitational force* rather than “weight”, to keep it distinct from what a scale reads. When you jump into the air you feel “weightless” because no normal force acts on you, even though the Earth still pulls you down. You have not turned gravity off.

A common misconception deserves to be addressed before we proceed. The normal force and the gravitational force are two completely different forces. They are equal in magnitude here only because the person is at rest, and this equality will fail the moment there is acceleration. More importantly, they do not form a Newton's Third Law interaction pair: both act on the same object (the person), whereas Third Law pairs always act on *different* objects. The true Third Law partner of the gravitational force on the person due to the Earth, $\vec{F}_{E,P}^g$, is the gravitational force on the Earth due to the person, $\vec{F}_{P,E}^g$, and they satisfy $\vec{F}_{E,P}^g = -\vec{F}_{P,E}^g$. These act on different objects: one on the person, the other on the Earth. Likewise, the Third Law partner of the normal force $\vec{\mathcal{N}}$ from the scale is the force the person's feet exert downward on the scale surface.

Experience in an Elevator

Now we take the bathroom scale somewhere interesting. If a scale measures the normal force — not gravity — then the reading should change whenever the normal force has to do something other than just hold you up. An elevator is the perfect laboratory: it can accelerate up, accelerate down, cruise, and (in a thought experiment we will keep hypothetical) drop freely.

²Newtonian mechanics treats mass as a fixed quantity. Special relativity shows that an object's inertia grows with its speed. One common way to express this introduces a velocity-dependent mass $m = m_0/\sqrt{1 - v^2/c^2}$, where m_0 is the rest mass and c is the speed of light; for everyday speeds $v \ll c$ this is indistinguishable from m_0 . Modern physics usually prefers instead to keep mass as the invariant m_0 and describe the growing inertia through momentum and energy. Either way, the effect is utterly negligible at ordinary speeds. We will return to this when we study modern physics.

Stop and Predict 5.2: Four rides on a scale

A 60 kg person stands on a scale in an elevator. At rest the scale reads 60.0 kg. Predict — more than 60, less than 60, exactly 60, or zero? — for each ride, *before* reading the analysis:

1. The elevator moves *upward at a steady* 5 m/s.
2. The elevator *accelerates upward*.
3. The elevator *accelerates downward* (slowing while going up, or speeding up while going down).
4. The cable snaps and the elevator *falls freely*.

Ride 1 is the one people argue about longest — “moving up” feels like it should read more. Commit to all four; the next pages check them one by one.

On the companion site: the **elevator simulator** (Chapter 5 on the course website) has an acceleration slider and a live scale readout — test your four predictions there before or after the analysis, and watch the cable snap in case 4.

In each case exactly two forces act on the person: the gravitational force $m\vec{g}$ directed downward and the normal force $\vec{\mathcal{N}}$ directed upward.

(a) At Rest or Moving at Constant Velocity.

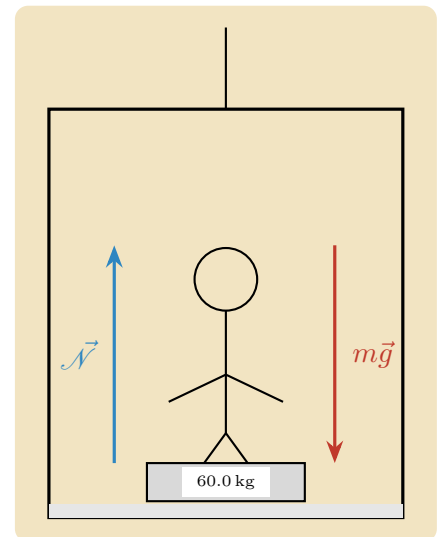
When the elevator is stationary or moving at constant velocity, the acceleration is zero. The person’s state of motion is not changing, so the floor need only support the person’s weight, nothing more. Newton’s second law in the vertical direction gives

$$\vec{\mathcal{N}} + \vec{F}_g = \vec{0} \implies \mathcal{N} - mg = 0,$$

so that

$$\mathcal{N} = mg = (60)(10) = 600 \text{ N}.$$

The scale displays $\mathcal{N}/g = 60.0 \text{ kg}$. Here the apparent weight equals the true weight, which is why we never notice the distinction in ordinary standing. Note what this settles about ride 1 of your predictions: a steady 5 m/s upward is *zero acceleration*, so the scale reads exactly 60.0 — the reading cares about acceleration, never about velocity. If a sealed elevator cruises perfectly smoothly, no scale (and no sensation in your stomach) can tell you whether you are moving at all. That is the first law speaking.



(b) Accelerating Upward.

Now the elevator accelerates upward. To make the person accelerate upward too, there must be a net upward force on them. But the person's inertia resists this change in motion: their body "wants" to stay at rest or moving as it was. The floor must therefore push up harder than before, hard enough both to support the weight and to supply the extra upward force the acceleration demands. The normal force rises above mg . Newton's second law gives

$$\mathcal{N} - mg = ma \implies \mathcal{N} = m(g + a). \quad (40)$$

Suppose the scale reads 64.8 kg, meaning $\mathcal{N} = 648 \text{ N}$:

$$648 = 60(10 + a) \implies a = 0.8 \text{ m s}^{-2} \text{ upward.}$$

The person feels heavier because the floor genuinely presses harder on their feet. The true weight is unchanged at 600 N; only the apparent weight has risen, to 648 N.

(c) Accelerating Downward.

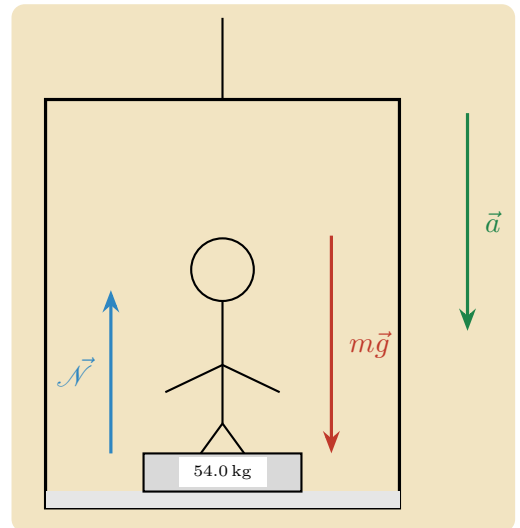
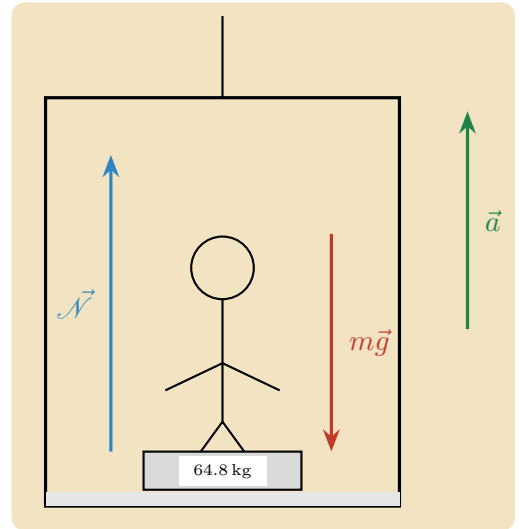
When the elevator accelerates downward, the person's inertia resists the change: the body "wants" to keep moving as it was, so for a moment it does not fall away with the floor as readily as the floor drops beneath it. The person presses less firmly on the scale, and so the scale presses less firmly back. The normal force falls below mg , leaving a net downward force that produces the downward acceleration:

$$mg - \mathcal{N} = ma \implies \mathcal{N} = m(g - a). \quad (41)$$

If the scale reads 54.0 kg, meaning $\mathcal{N} = 540 \text{ N}$:

$$540 = 60(10 - a) \implies a = 1.0 \text{ m s}^{-2} \text{ downward.}$$

The person feels lighter because the floor really does press less on their feet. The true weight is unchanged at 600 N; only the apparent weight has fallen, to 540 N.

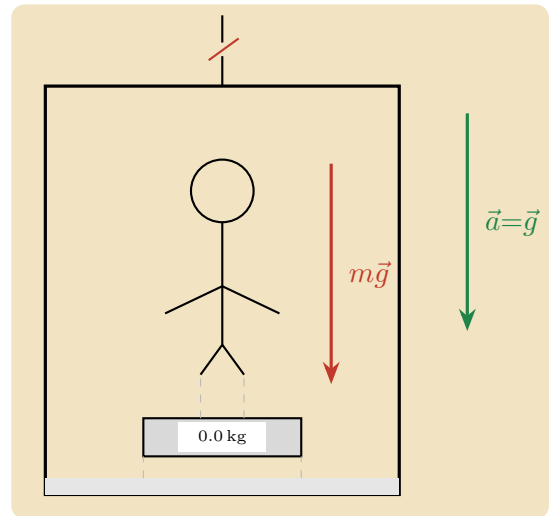


(d) Free Fall.

Now imagine the cable snaps and the elevator falls freely. Gravity acts on everything inside equally, so the person, the scale, and the floor all accelerate downward together at the same rate $a = g$. Because the floor falls away just as fast as the person does, it never has to push on them at all. The normal force drops all the way to zero. Setting $a = g$ in equation (41),

$$\mathcal{N} = m(g - g) = 0.$$

The scale reads zero, yet gravity has certainly not switched off; the Earth still pulls the person down with the full force mg . The person simply floats inside the falling car because there is no longer any contact force to register. This is exactly what we mean by *weightlessness*: not the absence of gravity, but the absence of the normal force.



Checkpoint 5.1: The floating astronaut

Astronauts on the International Space Station float freely inside the cabin. The station orbits about 400 km above the Earth's surface. The astronauts float because

- (a) at that altitude they are beyond the reach of the Earth's gravity.
- (b) the Earth's gravity is cancelled by the pull of the Moon and Sun.
- (c) they and the station are both in free fall, falling around the Earth together.
- (d) the station's speed generates an outward force that balances gravity.

Answer: (c). At 400 km up, the Earth's gravity is still about 90% of its surface strength — answer (a) is the single most common misconception in this course. The station and everyone in it are permanently in free fall, falling freely, together, all the time. Their sideways speed simply makes them keep missing the Earth — which is what an orbit is, as the section on gravitation will make precise.

Supplementary — A Glimpse of Something Deeper: The Equivalence Principle (optional — read for pleasure, skip without loss)

We close the elevator story with a question that troubled Einstein and ultimately led him to general relativity. Imagine you wake up inside a sealed box with no windows. You feel a normal force pressing up on your feet. How could you tell whether you are sitting stationary on the surface of the Earth, with gravity pulling you down, or drifting in empty space far from any planet, with the box accelerating upward at $g = 9.8 \text{ m/s}^2$?

The remarkable answer is that you cannot. No mechanical experiment conducted inside the box will distinguish between these two situations. Drop a ball: in both cases it accelerates toward the floor at 10 m/s^2 . Step on a scale: it reads mg in both

cases. Einstein elevated this to a fundamental principle: **gravity and accelerated motion are locally equivalent**. This is the **equivalence principle**.

a) Two Kinds of Mass. The symbol m appears in two logically distinct roles. In Newton's second law, m appears as **inertial mass**: the resistance of an object to acceleration. In the gravitational force law, m appears as **gravitational mass**: the quantity that determines how strongly gravity pulls on the object. There is no obvious reason these two quantities should be equal. Yet every experiment ever performed finds them equal. The Hungarian physicist Loránd Eötvös demonstrated this equality to one part in 10^8 in the 1880s, and modern experiments push this to one part in 10^{15} .

The equality $m_{\text{inertial}} = m_{\text{gravitational}}$ is exactly what ensures that all objects fall at the same rate regardless of their mass.

b) Light Must Bend. The equivalence principle has a striking consequence that goes beyond mechanics. Shine a horizontal beam of light across the accelerating box. While the light travels across, the box accelerates upward, so the beam hits the far wall slightly below where it was aimed — the light beam appears to curve downward. Since an accelerating box is locally indistinguishable from a gravitational field, gravity must also bend light. This was a radical prediction: Newtonian gravity acts only on objects with mass, and light was thought to be massless. Einstein's prediction was confirmed experimentally in 1919 by Arthur Eddington, who photographed stars near the edge of the Sun during a solar eclipse and found that their apparent positions were shifted by exactly the amount Einstein predicted. The announcement made Einstein famous overnight.

c) Where This Leads. Once you take the equivalence principle seriously and ask what geometry of space and time is consistent with it, you are led to **general relativity**. Gravity is reinterpreted not as a force but as the curvature of spacetime caused by mass and energy. Objects in free fall follow the straightest possible paths through this curved geometry. The bending of light, the slowing of clocks in gravitational fields, the existence of gravitational waves, and the physics of black holes all follow from the same geometric framework. The elevator we analysed today is the first step in that journey.

5.2 Circular Motion

We shall now investigate a special class of motions: motion in a plane about a central point, which we refer to as *central motion*. The most important special case is circular motion. There are many instances of circular motion: a bicycle rider on a circular track, a ball spun around on a string, and the rotation of a spinning wheel are just a few examples. The motion of the Moon around the Earth is nearly circular. The motions of the planets around the Sun are nearly circular. Our Sun moves in a nearly circular orbit about the centre of our galaxy.

We begin by describing the kinematics of circular motion – the position, velocity, and acceleration – as a special case of two-dimensional motion. Unlike linear motion, where velocity and acceleration are directed along the line of motion, in circular motion the velocity is always tangent to the circle. This means that as the object moves in a circle, the direction of the velocity is always changing. Examining this change reveals that the direction of the acceleration is towards the centre of the circle. This inward component of acceleration is called the *centripetal acceleration*. If the object is also changing speed, there is an additional non-zero tangential acceleration in the direction of motion. For uniform circular motion (constant speed), only the centripetal acceleration is non-zero.

Newton recognised that the Moon is always “falling” towards the Earth; otherwise, by the First Law, it would continue in a straight line rather than follow a circular orbit. There must therefore exist a centripetal force, a radial force pointing inward, producing the centripetal acceleration. Because Newton’s Second Law is a vector equation, the radial component gives

$$F_r = ma_r. \quad (42)$$

Circular Motion: Velocity and Angular Velocity

We describe circular motion using polar coordinates. The position vector $\vec{r}(t)$ of an object moving in a circular orbit of radius r is written as

$$\vec{r}(t) = r \hat{r}(t), \quad (43)$$

where $\hat{r}(t)$ is the radial unit vector pointing from the origin to the object. At any point P with coordinates $(r, \theta(t))$, we define two sets of unit vectors: the polar basis $(\hat{r}(t), \hat{\theta}(t))$ and the Cartesian basis $(\hat{i}, \hat{j})$, as shown in Figure 5.5. Their relationship is

$$\hat{r}(t) = \cos \theta(t) \hat{i} + \sin \theta(t) \hat{j}, \quad (44)$$

$$\hat{\theta}(t) = -\sin \theta(t) \hat{i} + \cos \theta(t) \hat{j}. \quad (45)$$

Before computing the velocity, we calculate the time derivatives of the unit vectors. Using the chain rule on Eq. (44),

$$\frac{d\hat{r}(t)}{dt} = (-\sin \theta(t) \hat{i} + \cos \theta(t) \hat{j}) \frac{d\theta}{dt} = \frac{d\theta}{dt} \hat{\theta}(t). \quad (46)$$

Similarly, differentiating Eq. (45),

$$\frac{d\hat{\theta}(t)}{dt} = (-\cos \theta(t) \hat{i} - \sin \theta(t) \hat{j}) \frac{d\theta}{dt} = -\frac{d\theta}{dt} \hat{r}(t). \quad (47)$$

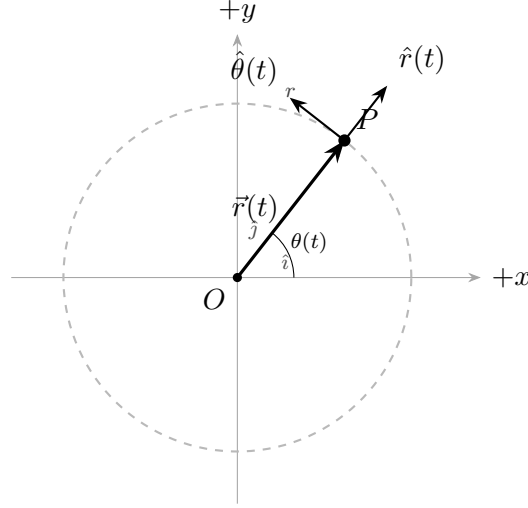


Figure 5.5: A circular orbit of radius r with the polar unit vectors $\hat{r}(t)$ and $\hat{\theta}(t)$ at the point P .

The velocity vector is then

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = r \frac{d\hat{r}}{dt} = r \frac{d\theta}{dt} \hat{\theta}(t) = v_{\theta} \hat{\theta}(t), \quad (48)$$

where the tangential component of the velocity is

$$v_{\theta} = r \frac{d\theta}{dt}. \quad (49)$$

The *angular speed* is defined as the magnitude of the rate of change of angle,

$$\omega \equiv \left| \frac{d\theta}{dt} \right|, \quad (50)$$

so that the speed is $v = |\vec{v}| = r\omega$.

Geometric Derivation of the Velocity

Consider a particle undergoing circular motion. During a time interval Δt , the particle moves from position $\vec{r}(t)$ to $\vec{r}(t + \Delta t)$, sweeping an angle $\Delta\theta$, as shown in Figure 5.6.

The magnitude of the displacement is $|\Delta\vec{r}| = 2r \sin(\Delta\theta/2)$. For small angles, $\sin(\Delta\theta/2) \approx \Delta\theta/2$, so

$$|\Delta\vec{r}| \approx r \Delta\theta. \quad (51)$$

This is consistent with the limit $\Delta\theta \rightarrow 0$, where the chord approaches the arc length $r \Delta\theta$. The magnitude of the velocity then follows as

$$v \equiv |\vec{v}(t)| = \lim_{\Delta t \rightarrow 0} \frac{|\Delta\vec{r}|}{\Delta t} = r \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = r \frac{d\theta}{dt} = r\omega. \quad (52)$$

As $\Delta t \rightarrow 0$, the direction of $\Delta\vec{r}$ approaches the tangent to the circle, perpendicular to $\vec{r}(t)$ and in the $+\hat{\theta}$ -direction for counterclockwise motion. This confirms that the velocity is always tangent to the circle.

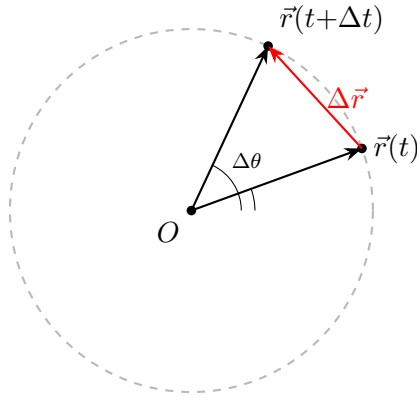


Figure 5.6: Displacement vector $\Delta \vec{r}$ for circular motion over an interval Δt .

Tangential and Radial Acceleration

In polar coordinates, the acceleration has two components: a tangential component a_θ and a radial component a_r ,

$$\vec{a}(t) = a_r \hat{r}(t) + a_\theta \hat{\theta}(t). \quad (53)$$

The unit vectors $\hat{r}(t)$ and $\hat{\theta}(t)$ change direction with time and are therefore not constant. To compute the acceleration, we differentiate the velocity $\vec{v}(t) = r(d\theta/dt) \hat{\theta}(t)$ using the product rule,

$$\vec{a}(t) = r \frac{d^2\theta}{dt^2} \hat{\theta}(t) + r \frac{d\theta}{dt} \frac{d\hat{\theta}}{dt}. \quad (54)$$

Substituting $d\hat{\theta}/dt = -(d\theta/dt)\hat{r}(t)$,

$$\vec{a}(t) = r \frac{d^2\theta}{dt^2} \hat{\theta}(t) - r \left(\frac{d\theta}{dt} \right)^2 \hat{r}(t). \quad (55)$$

The two components are therefore

$$a_\theta = r \frac{d^2\theta}{dt^2}, \quad (56)$$

$$a_r = -r \left(\frac{d\theta}{dt} \right)^2 = -r\omega^2 < 0. \quad (57)$$

Because $a_r < 0$, the radial component $a_r(t) = -r\omega^2 \hat{r}(t)$ is always directed towards the centre of the circular orbit. This is the centripetal acceleration.

Worked Example 5.2: Circular Motion Kinematics

A particle moves in a circle of radius R . At $t = 0$ it is on the x -axis. The angle with the positive x -axis is $\theta(t) = At^3 - Bt$, where A and B are positive constants. Find (a) the velocity vector and (b) the acceleration vector in polar coordinates. At what time is the centripetal acceleration zero?

Solution. The derivatives are $d\theta/dt = 3At^2 - B$ and $d^2\theta/dt^2 = 6At$. The velocity vector is

$$\vec{v}(t) = R \frac{d\theta}{dt} \hat{\theta}(t) = R(3At^2 - B) \hat{\theta}(t).$$

The acceleration vector is

$$\vec{a}(t) = R \frac{d^2\theta}{dt^2} \hat{\theta}(t) - R \left(\frac{d\theta}{dt} \right)^2 \hat{r}(t) = 6RA t \hat{\theta}(t) - R(3At^2 - B)^2 \hat{r}(t).$$

The centripetal acceleration is zero when $d\theta/dt = 0$, that is, when

$$3At_1^2 - B = 0 \implies t_1 = \sqrt{\frac{B}{3A}}.$$

Period and Frequency for Uniform Circular Motion

If the total tangential force on the object is zero, the tangential acceleration vanishes,

$$a_\theta = 0, \quad (58)$$

and the speed remains constant. This is *uniform circular motion*. The acceleration reduces to

$$\vec{a}_r(t) = -r\omega^2 \hat{r}(t) \quad (\text{uniform circular motion}). \quad (59)$$

Because the speed $v = r\omega$ is constant, the time to complete one full revolution, the *period* T , is constant. In one period the object travels the circumference $2\pi r$, so

$$2\pi r = vT \implies T = \frac{2\pi r}{v} = \frac{2\pi r}{r\omega} = \frac{2\pi}{\omega}. \quad (60)$$

The *frequency* f is the reciprocal of the period,

$$f = \frac{1}{T} = \frac{\omega}{2\pi}. \quad (61)$$

The SI unit of frequency is the hertz: $[\text{Hz}] = [\text{s}^{-1}]$. The magnitude of the centripetal acceleration can be written in several equivalent forms. Using $v = r\omega$,

$$|a_r| = r\omega^2, \quad (62)$$

$$|a_r| = \frac{v^2}{r}, \quad (63)$$

$$|a_r| = 4\pi^2 r f^2, \quad (64)$$

$$|a_r| = \frac{4\pi^2 r}{T^2}. \quad (65)$$

Which form to use depends on what information is given about the orbit.

Geometric Interpretation of the Radial Acceleration

An object in circular orbit is always accelerating towards the centre. The velocity is always tangent to the circle, so its direction is continuously changing as the object moves. Because the velocity changes direction, the acceleration is non-zero even when the speed is constant.

The magnitude of the change in velocity is $|\Delta \vec{v}| = 2v \sin(\Delta\theta/2) \approx v \Delta\theta$ for small $\Delta\theta$ (Figure 5.7). The magnitude of the radial acceleration is therefore

$$|a_r| = \lim_{\Delta t \rightarrow 0} \frac{|\Delta \vec{v}|}{\Delta t} = v \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = v \frac{d\theta}{dt} = v\omega = \frac{v^2}{r}. \quad (66)$$

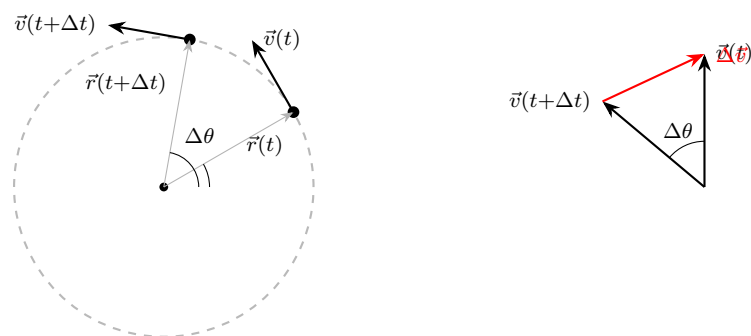


Figure 5.7: Left: velocity vectors at two nearby points on a circular orbit. Right: the change in velocity $\Delta\vec{v}$, which points inward as $\Delta\theta \rightarrow 0$.

As $\Delta\theta \rightarrow 0$, the direction of $\Delta\vec{v}$ is perpendicular to $\vec{v}$ and directed inward, in the $-\hat{r}$ -direction, confirming the centripetal character of the acceleration.

Angular Velocity and Angular Acceleration

Angular Velocity

We always choose a right-handed cylindrical coordinate system. With the positive z -axis pointing up, the angle θ increases counterclockwise as viewed from above. The *angular velocity vector* for a point object undergoing circular motion about the z -axis is directed along the z -axis,

$$\vec{\omega} = \frac{d\theta}{dt} \hat{k} = \omega_z \hat{k}. \quad (67)$$

The angular speed is the magnitude of the z -component, $\omega \equiv |\omega_z| = |d\theta/dt|$. If the object rotates counterclockwise, $\omega_z = d\theta/dt > 0$ and $\vec{\omega}$ points in the $+\hat{k}$ -direction. If the object rotates clockwise, $\omega_z < 0$ and $\vec{\omega}$ points in the $-\hat{k}$ -direction.

The velocity and angular velocity are related by the cross product,

$$\vec{v} = \vec{\omega} \times \vec{r} = \frac{d\theta}{dt} \hat{k} \times r\hat{r} = r \frac{d\theta}{dt} \hat{\theta}. \quad (68)$$

Worked Example 5.3: Angular Velocity

A particle moves in a circle of radius R . At $t = 0$ it is on the x -axis. The angle with the positive x -axis is $\theta(t) = At - Bt^3$, where A and B are positive constants. Find (a) the angular velocity vector, (b) the velocity vector, (c) the time t_1 when the angular velocity is zero, and (d) the direction of $\vec{\omega}$ for $t < t_1$ and $t > t_1$.

Solution. The derivative of $\theta(t)$ is

$$\frac{d\theta}{dt} = A - 3Bt^2.$$

(a) The angular velocity vector is

$$\vec{\omega}(t) = (A - 3Bt^2) \hat{k}.$$

(b) The velocity is

$$\vec{v}(t) = R(A - 3Bt^2) \hat{\theta}(t).$$

(c) The angular velocity is zero when

$$A - 3Bt_1^2 = 0 \implies t_1 = \sqrt{\frac{A}{3B}}.$$

(d) For $t < t_1$, $d\theta/dt > 0$ so $\vec{\omega}$ points in the $+\hat{k}$ -direction (counterclockwise). For $t > t_1$, $d\theta/dt < 0$ so $\vec{\omega}$ points in the $-\hat{k}$ -direction (clockwise).

Angular Acceleration

The *angular acceleration vector* for circular motion about the z -axis is

$$\vec{\alpha} = \frac{d^2\theta}{dt^2} \hat{k} = \alpha_z \hat{k}. \quad (69)$$

The SI units of angular acceleration are $[\text{rad} \cdot \text{s}^{-2}]$. There are four combinations of the signs of ω_z and α_z to consider.

When $\vec{\alpha}$ points in the $+\hat{k}$ -direction ($\alpha_z > 0$): (i) if $\omega_z > 0$ (counterclockwise), the object is speeding up counterclockwise; (ii) if $\omega_z < 0$ (clockwise), the object is slowing down.

When $\vec{\alpha}$ points in the $-\hat{k}$ -direction ($\alpha_z < 0$): (iii) if $\omega_z > 0$ (counterclockwise), the object is slowing down; (iv) if $\omega_z < 0$ (clockwise), the object is speeding up clockwise.

The tangential and radial components of the acceleration (Eqs. (56) and (57)) can be written compactly as $a_\theta = r\alpha_z$ and $a_r = -r\omega_z^2$.

Worked Example 5.4: Integration and Circular Motion Kinematics

A point object is constrained to travel in a circle. The z -component of the angular acceleration is

$$\alpha_z(t) = \begin{cases} b\left(1 - \frac{t}{t_1}\right), & 0 \leq t \leq t_1, \\ 0, & t > t_1, \end{cases}$$

where $b > 0$ has units $\text{rad} \cdot \text{s}^{-2}$. (a) Find the angular velocity at $t = t_1$. (b) Through what angle has the object rotated at $t = t_1$?

Solution. (a) Starting from rest ($\omega_z(0) = 0$), we integrate:

$$\omega_z(t_1) = \int_0^{t_1} b\left(1 - \frac{t'}{t_1}\right) dt' = b\left[t_1 - \frac{t_1}{2}\right] = \frac{bt_1}{2}.$$

(b) First find $\omega_z(t)$ for $0 \leq t \leq t_1$:

$$\omega_z(t) = \int_0^t b\left(1 - \frac{t'}{t_1}\right) dt' = b\left(t - \frac{t^2}{2t_1}\right).$$

Then integrate again to find the angle:

$$\theta(t_1) - \theta(0) = \int_0^{t_1} \omega_z(t') dt' = \int_0^{t_1} b\left(t' - \frac{t'^2}{2t_1}\right) dt' = b\left[\frac{t_1^2}{2} - \frac{t_1^2}{6}\right] = \frac{bt_1^2}{3}.$$

5.3 Gravitation: The Moon is Falling

At the start of our circular-motion story we repeated Newton’s remark: the Moon is always *falling* towards the Earth — otherwise, by the first law, it would fly off in a straight line. We now have every tool needed to take that remark seriously, and it leads to one of the greatest predict-and-check episodes in the history of science. This section retells it, and asks you to redo Newton’s arithmetic yourself.

In 1665 the plague closed Cambridge University, and the 23-year-old Newton retreated to his family farm at Woolsthorpe. By his own (much later) telling, watching an apple fall in the garden set him wondering: the Earth’s pull evidently reaches the top of the tallest tree — might it reach all the way to the Moon? If so, the Moon’s endless circling and the apple’s brief drop would be *the same phenomenon*. No one had ever connected the heavens and the ground by a single law. The idea is bold; what makes it science is that it can be checked by a number.

Stop and Predict 5.3: Newton’s own prediction

Suppose the Earth’s pull weakens with the *square* of the distance from the Earth’s centre — Newton’s guess, partly inspired by how the light of a lamp dilutes with distance. An apple sits about one Earth radius from the centre ($R_E \approx 6.37 \times 10^6$ m). The Moon sits about *sixty* Earth radii away. Under the inverse-square guess:

1. How many times weaker should gravity be at the Moon than at the apple?
2. The apple accelerates at $g = 9.8$ m/s². Predict the Moon’s acceleration.

Answer both before reading on — then we will *measure* the Moon’s actual acceleration, using nothing but the circular-motion formulas you just learned and two numbers astronomers have known since antiquity.

The Moon Test

If the inverse-square guess is right, gravity at the Moon should be weaker by a factor of $60^2 = 3600$, so the Moon should be falling with acceleration

$$a_{\text{predicted}} = \frac{g}{3600} = \frac{9.8}{3600} \approx 2.7 \times 10^{-3} \text{ m/s}^2.$$

That is the prediction. Now the check. The Moon moves in a (nearly) circular orbit, and for circular motion we know exactly what the centripetal acceleration must be — Eq. (65):

$$|a_r| = \frac{4\pi^2 r}{T^2}.$$

The orbital radius is $r = 3.84 \times 10^8$ m and the orbital period is $T = 27.3$ days = 2.36×10^6 s. Therefore the Moon’s actual acceleration is

$$|a_r| = \frac{4\pi^2(3.84 \times 10^8)}{(2.36 \times 10^6)^2} \approx 2.7 \times 10^{-3} \text{ m/s}^2.$$

The prediction and the measurement agree. Newton, recalling the calculation years later, wrote that he “compared the force requisite to keep the Moon in her orb with

the force of gravity at the surface of the earth, and found them answer pretty nearly.” *Pretty nearly* — the understatement of the seventeenth century. The apple and the Moon fall by the same law, and the law is inverse-square.

Notice what kind of argument this is: guess a law, extract a numerical prediction, and confront it with a measurement that had no right to agree by accident. A factor of 3600, predicted from a hunch about lamplight, confirmed to within a percent. You will use exactly this pattern — predict, then check — in your own experiments this year, if with slightly smaller stakes.

The Law of Universal Gravitation

Newton generalised the moon test into a law that applies to every pair of masses in the universe — two galaxies, or two sesame seeds.

Newton’s Law of Universal Gravitation

Any two point masses m_1 and m_2 separated by a distance r attract each other along the line joining them with a force of magnitude

$$F_g = G \frac{m_1 m_2}{r^2}, \quad (70)$$

where G is the **universal gravitational constant**,

$$G = 6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}.$$

By the Third Law the two masses feel equal and opposite pulls: the Earth pulls the apple, and the apple pulls the Earth, with the same force.

Two remarks. First, for a sphere (like the Earth) the law applies as if all the mass were concentrated at the centre — a theorem that cost Newton real effort to prove, and part of why he delayed publishing for twenty years. That is why the apple’s distance counts as one Earth *radius*, measured from the centre, not from the ground. Second, the constant G is astonishingly small: gravity is by far the weakest force you will meet in this course. Two 1 kg masses held 1 m apart attract with about 10^{-10} N — roughly the weight of a single flake of dust. Gravity runs the universe only because masses like planets are enormous, and because, unlike electricity, gravity has no negative charge to cancel it.

The value of G was not measured in Newton’s lifetime. In 1798 Henry Cavendish — a man so shy he communicated with his housekeeper by notes — suspended a dumbbell of small lead spheres from a fine wire and measured the twist produced by the gravitational pull of two larger spheres, a whisper of a force in a sealed room. Knowing G , one can put the Earth itself on the balance: from g and R_E , the Earth’s mass follows at once (try the next example). Cavendish’s contemporaries called his experiment “weighing the Earth.”

Worked Example 5.5: Weighing the Earth

Given $g = 9.81 \text{ m/s}^2$ at the surface, $R_E = 6.37 \times 10^6 \text{ m}$, and Cavendish's G , find the mass of the Earth.

Solution. A mass m at the surface feels the attraction of the whole Earth acting from its centre:

$$mg = G \frac{M_E m}{R_E^2} \implies M_E = \frac{gR_E^2}{G} = \frac{(9.81)(6.37 \times 10^6)^2}{6.674 \times 10^{-11}} \approx 5.97 \times 10^{24} \text{ kg}.$$

Notice that m cancelled: *every* mass at the surface accelerates at the same $g = GM_E/R_E^2$, which is the deep reason (first noticed by Galileo) that heavy and light objects fall together. Notice also what the formula says about altitude: at the International Space Station, $r = R_E + 400 \text{ km}$, and g drops only to about 8.7 m/s^2 — 89% of its surface value, exactly as Checkpoint 5.1 claimed.

Circular Orbits

For a satellite of mass m in a circular orbit of radius r about the Earth, gravity is the only force, and it must supply exactly the centripetal acceleration v^2/r that circular motion demands. Newton's second law in the radial direction gives

$$G \frac{M_E m}{r^2} = \frac{mv^2}{r} \implies v = \sqrt{\frac{GM_E}{r}}. \quad (71)$$

The satellite's mass cancels again — a heavy satellite and a paint flake share any given orbit at the same speed. Writing $v = 2\pi r/T$ and squaring:

$$T^2 = \frac{4\pi^2}{GM_E} r^3. \quad (72)$$

The square of the period is proportional to the *cube* of the orbital radius. Keep this result in view; it is about to connect us to an astronomer who died twelve years before Newton was born.

Kepler's Laws

Between 1576 and 1601, on an island observatory financed by the Danish crown, Tycho Brahe measured the positions of the planets night after night for twenty-five years — the finest naked-eye data ever taken, accurate to about one-sixtieth of a degree. Johannes Kepler, his difficult and brilliant assistant, inherited the data and spent nearly a decade in “war with Mars,” trying orbit after orbit until the numbers finally surrendered. What they surrendered to are three laws, published in 1609 and 1619. We state them here; deriving them fully belongs to a later chapter, but you should know what they say and where they came from.

Kepler's Laws of Planetary Motion

1. **The Law of Ellipses.** Each planet moves on an *ellipse* with the Sun at one focus. (A circle is the special case of an ellipse with both foci together.)

2. **The Law of Equal Areas.** The line from the Sun to a planet sweeps out equal areas in equal times — so a planet moves fastest when nearest the Sun and slowest when farthest.
3. **The Harmonic Law.** The square of a planet’s orbital period is proportional to the cube of its mean distance from the Sun: $T^2 \propto a^3$, with the *same* constant of proportionality for every planet.

Kepler found these laws empirically — pure pattern, no mechanism. He never knew *why* they were true, though he suspected the Sun. Now look back at Eq. (72): for circular orbits, the third law fell out of the law of gravitation in two lines of algebra, complete with the constant $4\pi^2/GM$ — the same for every body orbiting the same central mass, just as Kepler said. (Done with ellipses instead of circles, the same follows exactly; that computation is beyond us this year.) This was the argument that convinced Newton’s contemporaries: one law, guessed from a falling apple, reproduced three laws distilled from twenty-five years of Tycho’s data. The second law, the equal-areas rule, has an even prettier origin — it turns out to be a disguised conservation law, and it will fall into our hands almost for free when we meet angular momentum in the next chapter.

Whiteboard Challenge 5.1: The satellite that stands still (*stand up, groups of three, one marker each*)

Television dishes in Bhutan point at a fixed spot in the sky. The satellite they aim at is *geostationary*: it hangs over the same point on the equator forever.

- (a) Before computing: what must such a satellite’s orbital period be?
- (b) Use Eq. (72) to find the orbital radius, and hence the altitude above the ground. ($M_E = 5.97 \times 10^{24}$ kg; the sidereal day is $T = 86\,164$ s.)
- (c) A minister asks for a geostationary satellite parked directly above Paro (latitude 27°N), to improve coverage. As a group, decide: is this possible? Defend your answer with the physics of circular motion — where must the centre of any Earth orbit lie?

(You should find $r \approx 4.2 \times 10^4$ km, an altitude of about 36 000 km — a tenth of the way to the Moon. And no: the centre of every orbit must be the Earth’s centre, so a satellite hovering off the equator is impossible; it would need a circle whose centre is inside northern India.)

Supplementary — The Tides, or: the Sea Breathes Twice a Day (*optional — read for pleasure, skip without loss*)

Bhutan is landlocked, so here is something many of us have never watched: at the ocean’s edge, the water level rises and falls by a metre or more, twice every day. Newton’s law explains this breathing — and the explanation is subtler than “the Moon pulls the water.”

If the Moon simply pulled the water, there would be *one* bulge of ocean on the Moon’s side, and one high tide per day as the Earth turns through it. The observed

rhythm — two high tides a day — says otherwise. The key is that the Moon's pull is *inverse-square*, so it is not uniform across the Earth: the near-side ocean (about $59 R_E$ from the Moon) is pulled harder than the Earth's centre ($60 R_E$), which in turn is pulled harder than the far-side ocean ($61 R_E$). Relative to the Earth, the near water is pulled ahead, and the Earth is pulled away from the far water. Result: *two* bulges, one facing the Moon and one facing away, and every coast passes through both each day. (The Sun pulls far harder in absolute terms, but tides care about the *difference* in pull across the Earth, and by that measure the nearby Moon wins — the Sun's tide is about half the Moon's. When Sun and Moon line up, at new and full moon, the tides run extra high.)

How big is the effect? In mid-ocean, under a metre. But geography can amplify it spectacularly. The world record is the Bay of Fundy in eastern Canada, where the water rises and falls up to *sixteen metres* — a four-storey building, twice daily. The bay is a long, tapering funnel whose natural sloshing period happens to be close to the 12.4-hour tidal rhythm, so each tide arrives in step with the bay's own oscillation and pumps it higher, the way well-timed pushes build up a playground swing. Resonance — an idea that will return, from swings to musical instruments to circuits.

Newton's tidal theory, published in the *Principia* in 1687, was the first time in history the tides had an explanation rather than a mere timetable. Same law, third appearance: the apple, the Moon, and the sea.

Work, Energy, and Momentum

6.1 Introduction: The Grand Conservation Laws

In the previous chapters, we analyzed motion by looking at forces moment by moment. We asked: what is the acceleration right now? Where will the particle be in the next instant? This is Newton’s direct path, and it is incredibly powerful. But as systems grow from a single particle to many interacting pieces, tracking every force at every instant becomes a nightmare.

Fortunately, nature has hidden shortcuts. If we step back and look at the beginning and the end of a process, we discover that certain quantities remain completely unchanged, no matter how chaotic the middle was. These are the **conservation laws**. In this chapter, we will meet three of them: the conservation of linear momentum, the conservation of energy, and the conservation of angular momentum. Together, they form the backbone of classical mechanics and provide a modern, first-principles way to understand everything from a colliding billiard ball to a launching rocket and an orbiting planet.

6.2 Momentum and Its Conservation

Let us begin with a simple observation. A cricket ball and a tennis ball are thrown at you at the same speed. Which is harder to catch? The cricket ball, of course. Now imagine a tennis ball thrown at a gentle pace versus one served by a professional at 200 km/h. Again, the faster one is harder to catch.

What makes an object hard to stop is a combination of its mass and its velocity. Newton called this the “quantity of motion.” Today, we call it **linear momentum**, denoted by the vector $\vec{p}$.

Linear Momentum

The **linear momentum** $\vec{p}$ of a particle of mass m moving with velocity $\vec{v}$ is the product of its mass and its velocity:

$$\vec{p} = m\vec{v}. \quad (73)$$

Momentum is a vector quantity. Its direction is the same as the velocity, and its SI unit is the kilogram-metre per second (kg m/s).

Newton's second law was originally written not as $\vec{F} = m\vec{a}$, but in terms of momentum:

$$\vec{F} = \frac{d\vec{p}}{dt}. \quad (74)$$

If mass is constant, this recovers $\vec{F} = d(m\vec{v})/dt = m(d\vec{v}/dt) = m\vec{a}$. But Eq. (74) is more fundamental because it holds even when mass is changing (like a rocket burning fuel) and in the relativistic domain where mass appears to increase with speed.

Impulse and the Change in Momentum

If we multiply a force by the short time interval dt over which it acts, we get a quantity called **impulse**, $\vec{J}$:

$$d\vec{J} = \vec{F} dt \implies \vec{J} = \int_{t_1}^{t_2} \vec{F} dt. \quad (75)$$

Integrating Eq. (74) from t_1 to t_2 , we obtain the **impulse-momentum theorem**:

$$\vec{J} = \vec{p}_2 - \vec{p}_1 = \Delta\vec{p}. \quad (76)$$

The impulse applied to an object is exactly equal to its change in momentum. This explains why safety nets, crumple zones in cars, and catching a cricket ball with “soft hands” (pulling your hands back as you catch) reduce the impact force. By increasing the time interval Δt over which the momentum change occurs, the average force is reduced: $\vec{F}_{\text{avg}} = \Delta\vec{p}/\Delta t$.

Worked Example The Soft Hands of a Fielder

A cricket ball of mass $m = 0.16 \text{ kg}$ reaches a fielder traveling at $v = 20 \text{ m/s}$.

- What is the initial momentum of the ball?
- If the fielder catches the ball by holding their hands rigid, bringing it to rest in $\Delta t = 0.01 \text{ s}$, what is the average force felt by their hands?
- If they pull their hands back during the catch, extending the time to $\Delta t = 0.15 \text{ s}$, what is the average force?

Solution:

- $p_i = mv = (0.16 \text{ kg})(20 \text{ m/s}) = 3.2 \text{ kg m/s}$.
- The change in momentum is $\Delta p = p_f - p_i = 0 - 3.2 = -3.2 \text{ kg m/s}$. The average force is:

$$F_{\text{avg}} = \frac{\Delta p}{\Delta t} = \frac{-3.2 \text{ kg m/s}}{0.01 \text{ s}} = -320 \text{ N}.$$

This is about 32 kg of weight slamming into the hands.

- With the extended time:

$$F_{\text{avg}} = \frac{-3.2 \text{ kg m/s}}{0.15 \text{ s}} \approx -21.3 \text{ N}.$$

Pulling the hands back reduces the force by a factor of 15, making the catch safe and painless.

Conservation of Linear Momentum

Consider a system of two colliding particles, A and B . During the collision, particle A exerts a force $\vec{F}_{B \rightarrow A}$ on B , and by Newton's third law, B exerts an equal and opposite force $\vec{F}_{A \rightarrow B} = -\vec{F}_{B \rightarrow A}$ on A . If no external forces act on the system:

$$\frac{d\vec{p}_A}{dt} + \frac{d\vec{p}_B}{dt} = \vec{F}_{B \rightarrow A} + \vec{F}_{A \rightarrow B} = \vec{0} \implies \frac{d}{dt}(\vec{p}_A + \vec{p}_B) = \vec{0}. \quad (77)$$

Thus, the total momentum $\vec{P}_{\text{total}} = \vec{p}_A + \vec{p}_B$ is constant.

Conservation of Linear Momentum

If the net external force acting on a system of particles is zero, the total linear momentum of the system remains constant:

$$\vec{P}_{\text{initial}} = \vec{P}_{\text{final}}. \quad (78)$$

Stop and Predict 6.1: The Recoil of a Rifle

A heavy rifle of mass M fires a light bullet of mass m horizontally with velocity $\vec{v}_b$ relative to the ground. Before firing, everything is at rest. Predict: which direction will the rifle move when fired, and will its speed be greater than, equal to, or less than the bullet's speed? Commit to your prediction before reading further.

Since the net external horizontal force on the rifle-bullet system is zero, momentum must be conserved:

$$\vec{P}_i = \vec{0} \implies \vec{P}_f = M\vec{v}_r + m\vec{v}_b = \vec{0} \implies \vec{v}_r = -\frac{m}{M}\vec{v}_b. \quad (79)$$

The negative sign means the rifle moves in the opposite direction (recoil). Since the rifle is much heavier than the bullet ($M \gg m$), its speed is much smaller ($v_r \ll v_b$).

Rocket Propulsion: The Tsiolkovsky Equation

A rocket is a beautiful application of momentum conservation in which the mass of the system changes continuously as fuel is burned and ejected.

Let a rocket of mass m be moving with speed v at time t . In a short time interval dt , it ejects a mass dm of gas backward with an exhaust speed u relative to the rocket (which means speed $v - u$ relative to the ground). The rocket's mass becomes $m - dm$, and its speed increases to $v + dv$.

Applying momentum conservation in the ground frame (ignoring gravity for now):

$$P(t) = P(t + dt) \implies mv = (m - dm)(v + dv) + dm(v - u). \quad (80)$$

Expanding this and ignoring the tiny second-order product $dm dv$:

$$mv = mv + m dv - v dm - u dm + v dm \implies m dv = u dm. \quad (81)$$

Since the rocket's mass decreases, the change in the rocket's mass is $dm = -dm_{\text{rocket}}$. Let us write this as $m dv = -u dm$, where dm is now the differential change in the

rocket's own mass ($dm < 0$). Dividing by m and integrating:

$$\int_{v_0}^v dv' = -u \int_{m_0}^m \frac{dm'}{m'} \implies v - v_0 = u \ln \left(\frac{m_0}{m} \right). \quad (82)$$

This is the famous **Tsiolkovsky rocket equation**, derived independently by the Russian schoolteacher Konstantin Tsiolkovsky in 1897. It shows that to reach high speeds, a rocket must either have an extremely high exhaust velocity u or eject a massive fraction of its initial mass as fuel ($m_0/m \gg 1$).

6.3 Work and Kinetic Energy

We now shift to our second conservation law: energy. Historically, the concept of energy was much slower to develop than momentum. While Newton's mechanics focused on forces and momentum, Leibniz and others argued that another quantity, which they called *vis viva* ("living force") and computed as mv^2 , was also conserved in certain collisions. This was the precursor to kinetic energy.

To connect forces to energy, we define the concept of **work**.

Work Done by a Force

If a constant force $\vec{F}$ acts on an object while it undergoes a displacement $\vec{d}$, the work W done by the force is:

$$W = \vec{F} \cdot \vec{d} = Fd \cos \phi, \quad (83)$$

where ϕ is the angle between the force and displacement vectors. If the force varies along the path, we must integrate:

$$W = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r}. \quad (84)$$

Work is a scalar quantity. Its SI unit is the newton-metre, which we rename the **joule** (J) in honour of James Prescott Joule.

The Work-Kinetic Energy Theorem

What does work do? Let us compute the work done by a net force $\vec{F}_{\text{net}}$ on a particle of mass m as it moves along a path:

$$W = \int_1^2 \vec{F}_{\text{net}} \cdot d\vec{r} = \int_1^2 m \frac{d\vec{v}}{dt} \cdot d\vec{r}. \quad (85)$$

Since $\vec{v} = d\vec{r}/dt$, we can swap the variables using the chain rule: $d\vec{r} = \vec{v} dt$. Thus:

$$W = \int_{t_1}^{t_2} m \frac{d\vec{v}}{dt} \cdot \vec{v} dt = \int_{v_1}^{v_2} m \vec{v} \cdot d\vec{v}. \quad (86)$$

Since $\vec{v} \cdot d\vec{v} = v dv$ (the change in speed squared), we get:

$$W = \int_{v_1}^{v_2} mv dv = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2. \quad (87)$$

Work-Kinetic Energy Theorem

The net work done on a particle is exactly equal to the change in its **kinetic energy** K :

$$W_{\text{net}} = \Delta K, \quad \text{where } K \equiv \frac{1}{2}mv^2. \quad (88)$$

This theorem connects kinematic changes in speed directly to the forces acting over distances.

6.4 Potential Energy and Conservative Forces

Some forces, like gravity or a spring, are special: the work they do on an object as it moves from point A to point B depends only on the endpoints A and B , not on the path taken. Such forces are called **conservative**.

Conservative Forces

A force $\vec{F}$ is **conservative** if either of these equivalent statements holds:

1. The work done by $\vec{F}$ around any closed loop is zero: $\oint \vec{F} \cdot d\vec{r} = 0$.
2. The work done by $\vec{F}$ between two points is independent of the path connecting them.

For any conservative force, we can define a scalar function called **potential energy**, $U(\vec{r})$, such that the work done by the force is the negative change in this potential energy:

$$W_{\text{cons}} = -\Delta U = U_1 - U_2. \quad (89)$$

In one dimension, this relationship is:

$$F_x = -\frac{dU}{dx}. \quad (90)$$

For gravity near the Earth's surface, $U(y) = mgy$ (where $F_y = -mg$). For a spring obeying Hooke's law ($F_x = -kx$), $U(x) = \frac{1}{2}kx^2$.

Conservation of Mechanical Energy

If only conservative forces do work on a system, then $W_{\text{net}} = W_{\text{cons}} = -\Delta U$. But by the work-KE theorem, $W_{\text{net}} = \Delta K$. Equating these:

$$\Delta K = -\Delta U \implies \Delta(K + U) = 0. \quad (91)$$

We define the **total mechanical energy** as $E \equiv K + U$. Thus, if no non-conservative forces (like friction or drag) do work:

$$E = K + U = \text{constant}. \quad (92)$$

If non-conservative forces *do* work (like friction f_k), we write:

$$W_{\text{net}} = W_{\text{cons}} + W_{\text{nc}} = -\Delta U + W_{\text{nc}} = \Delta K \implies E_2 - E_1 = W_{\text{nc}}. \quad (93)$$

Since friction always does negative work, mechanical energy is lost to heat (thermal energy). The total energy of the universe is still conserved, but mechanical energy is dissipated.

Worked Example The Loop-the-Loop

A small block of mass m slides down a frictionless track and enters a circular loop of radius R . What is the minimum height h from which the block must start (from rest) to make it all the way around the loop without falling off at the top?

Solution: Let us first look at the top of the loop (point T , height $2R$). Two forces act on the block: gravity mg downward and the normal force $\mathcal{N}$ downward. For circular motion:

$$\Sigma F_{\text{radial}} = mg + \mathcal{N} = m \frac{v_T^2}{R}.$$

To just maintain contact, the normal force must not drop below zero ($\mathcal{N} \geq 0$). Thus, the minimum speed required at the top is:

$$v_T^2 \geq gR.$$

Now we use conservation of mechanical energy between the starting point (height h , speed $v_0 = 0$) and the top (height $2R$, speed v_T):

$$E_{\text{initial}} = E_{\text{final}} \\ mgh + 0 = mg(2R) + \frac{1}{2}mv_T^2.$$

Canceling m and substituting the threshold speed $v_T^2 = gR$:

$$gh = 2gR + \frac{1}{2}gR \implies h = \frac{5}{2}R.$$

To clear the loop, the block must start at a height at least 2.5 times the radius of the loop.

Checkpoint 6.2

A cart is released from rest at height h on a roller coaster track. It goes down a valley and up a hill of height $H < h$. If friction is present:

- A. The speed at the top of the hill is $\sqrt{2g(h-H)}$. B. The speed is less than $\sqrt{2g(h-H)}$.
C. The speed is greater than $\sqrt{2g(h-H)}$. D. The cart cannot reach the hill.

B. Because friction does negative work, some mechanical energy is lost to thermal energy, so the speed at the top of the hill will be less than the frictionless prediction.

6.5 Collisions in 2D

Let us bring momentum and energy together to analyze collisions. In any collision where the net external force is zero, ****momentum is conserved****. However, kinetic energy is not always conserved: - An ****elastic collision**** is one in which the total kinetic energy is conserved: $K_i = K_f$. - An ****inelastic collision**** is one in which kinetic

energy is not conserved: $K_i > K_f$ (energy is lost to sound, deformation, and heat). If the objects stick together, the collision is **perfectly inelastic**.

In two dimensions, we write momentum conservation as vector components:

$$P_{i,x} = P_{f,x} \quad (94)$$

$$P_{i,y} = P_{f,y}. \quad (95)$$

Worked Example 2D Elastic Collision of Identical Balls

A billiard ball A of mass m moving with speed v along the x -axis collides elastically with an identical ball B of mass m at rest. After the collision, ball A moves off at angle θ_A relative to the x -axis, and ball B moves off at angle θ_B . Prove that the angle between their final paths, $\theta_A + \theta_B$, is exactly 90° (unless the collision is head-on).

Solution: Let $\vec{v}$ be the initial velocity of ball A . Let $\vec{v}_A$ and $\vec{v}_B$ be the final velocities of the two balls. Conservation of linear momentum gives:

$$m\vec{v} = m\vec{v}_A + m\vec{v}_B \implies \vec{v} = \vec{v}_A + \vec{v}_B.$$

Let us take the dot product of this vector equation with itself:

$$v^2 = (\vec{v}_A + \vec{v}_B) \cdot (\vec{v}_A + \vec{v}_B) = v_A^2 + v_B^2 + 2\vec{v}_A \cdot \vec{v}_B. \quad (96)$$

Because the collision is elastic, kinetic energy is also conserved:

$$\frac{1}{2}mv^2 = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 \implies v^2 = v_A^2 + v_B^2.$$

Subtracting this from Eq. (96), we find:

$$2\vec{v}_A \cdot \vec{v}_B = 0 \implies \vec{v}_A \cdot \vec{v}_B = 0.$$

If the dot product of the final velocity vectors is zero, they must be perpendicular! Thus, the angle between them is $\theta_A + \theta_B = 90^\circ$. (This holds for any non-head-on elastic collision of identical masses).

6.6 The Center of Mass

If we drop a spinning, vibrating wrench, its motion looks incredibly complex. But if we paint a bright dot at one specific point on the wrench, we discover that the dot traces out a perfect parabola, exactly like a single point particle. That special point is the **center of mass**.

Center of Mass

For a system of N discrete particles with masses m_i and positions $\vec{r}_i$, the position of the center of mass $\vec{R}_{\text{cm}}$ is:

$$\vec{R}_{\text{cm}} = \frac{1}{M} \sum_{i=1}^N m_i \vec{r}_i, \quad \text{where } M \equiv \sum_{i=1}^N m_i. \quad (97)$$

For a continuous object of mass density ρ , we integrate:

$$\vec{R}_{\text{cm}} = \frac{1}{M} \int \vec{r} dm. \quad (98)$$

Differentiating the position of the center of mass with respect to time, we find:

$$\vec{V}_{\text{cm}} = \frac{d\vec{R}_{\text{cm}}}{dt} = \frac{1}{M} \sum m_i \vec{v}_i = \frac{\vec{P}_{\text{total}}}{M}. \quad (99)$$

Thus, the total momentum of a system is simply the total mass multiplied by the velocity of the center of mass: $\vec{P}_{\text{total}} = M\vec{V}_{\text{cm}}$. If no net external force acts, $\vec{P}_{\text{total}}$ is constant, which means the center of mass moves at constant velocity, completely unaffected by any internal collisions or explosions!

6.7 Angular Momentum and Torque

Just as linear momentum describes an object's resistance to stopping its straight-line motion, ****angular momentum**** describes its resistance to stopping its rotation.

For a single particle of mass m at position $\vec{r}$ relative to an origin, moving with momentum $\vec{p} = m\vec{v}$, we define its angular momentum $\vec{L}$ relative to that origin using the cross product:

Angular Momentum of a Particle

The ****angular momentum**** $\vec{L}$ of a particle is:

$$\vec{L} \equiv \vec{r} \times \vec{p}. \quad (100)$$

Its magnitude is $L = rp \sin \theta$, where θ is the angle between $\vec{r}$ and $\vec{p}$. The units of angular momentum are $\text{kg m}^2/\text{s}$.

What changes angular momentum? Let us take the time derivative of $\vec{L}$:

$$\frac{d\vec{L}}{dt} = \frac{d}{dt}(\vec{r} \times \vec{p}) = \left(\frac{d\vec{r}}{dt} \times \vec{p} \right) + \left(\vec{r} \times \frac{d\vec{p}}{dt} \right). \quad (101)$$

The first term is $\vec{v} \times (m\vec{v}) = m(\vec{v} \times \vec{v}) = \vec{0}$, since the cross product of any vector with itself is zero. The second term is $\vec{r} \times \vec{F}_{\text{net}}$, using Newton's second law. We call the cross product of position and force the ****torque****, $\vec{\tau}$:

$$\vec{\tau} \equiv \vec{r} \times \vec{F}. \quad (102)$$

Thus, we arrive at the rotational version of Newton's second law:

$$\vec{\tau}_{\text{net}} = \frac{d\vec{L}}{dt}. \quad (103)$$

If the net torque on a system is zero, its angular momentum is conserved: $\vec{L}_{\text{initial}} = \vec{L}_{\text{final}}$.

Kepler's Second Law: A Disguised Conservation Law

We promised in Chapter 5 that Kepler's second law — that a planet's position vector sweeps out equal areas in equal times — is actually just the conservation of angular momentum in disguise. Let us prove it.

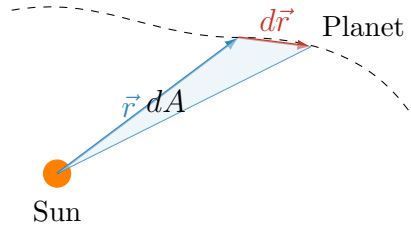


Figure 6.1: The area dA swept out by a planet's radius vector $\vec{r}$ in time dt is a triangle defined by $\vec{r}$ and the displacement $d\vec{r}$.

The area of the tiny triangle swept out by the radius vector $\vec{r}$ as the planet moves through displacement $d\vec{r} = \vec{v} dt$ is half the area of the parallelogram formed by $\vec{r}$ and $d\vec{r}$:

$$dA = \frac{1}{2} |\vec{r} \times d\vec{r}| = \frac{1}{2} |\vec{r} \times \vec{v}| dt. \quad (104)$$

Dividing by dt , the rate at which area is swept out (the areal velocity) is:

$$\frac{dA}{dt} = \frac{1}{2} |\vec{r} \times \vec{v}|. \quad (105)$$

Since $\vec{p} = m\vec{v}$, we can write $\vec{r} \times \vec{v} = (\vec{r} \times \vec{p})/m = \vec{L}/m$. Thus:

$$\frac{dA}{dt} = \frac{|\vec{L}|}{2m}. \quad (106)$$

Now, what is the torque acting on the planet? Gravity points directly along the line connecting the planet and the Sun. Thus, the force $\vec{F}_g$ is parallel to the radius vector $\vec{r}$. The torque is:

$$\vec{\tau} = \vec{r} \times \vec{F}_g = \vec{0}. \quad (107)$$

Since the torque is zero, the planet's angular momentum $\vec{L}$ is constant! Since L and the mass m are constant, Eq. (106) tells us that dA/dt is constant.

The planet sweeps out equal areas in equal times because gravity is a **central force** (pointing directly toward the source), which ensures that the planet's angular momentum remains perfectly conserved. What seemed a wonder is no wonder.

Whiteboard Challenge 6.1: Spinning in a Chair (*stand up, groups of three, one marker each*)

Sit in a swivel office chair that spins freely. Hold two heavy books in your hands and stretch your arms out wide. Have a partner spin you gently. Now, pull your arms in tight to your chest.

- (a) Predict: what will happen to your spinning speed when you pull your arms in?
- (b) Why? Use the conservation of angular momentum. Identify what happens to your distribution of mass (moment of inertia) and how that affects your angular speed ω .
- (c) If you extend your arms again, does your speed return to its original value?

6.8 Summary of Conservation Laws

The Three Pillars of Mechanics

- **Linear Momentum**: $\vec{p} = m\vec{v}$. Conserved if net external force is zero: $\Sigma \vec{F}_{\text{ext}} = \vec{0}$.
- **Mechanical Energy**: $E = K + U$, where $K = \frac{1}{2}mv^2$. Conserved if only conservative forces do work: $W_{\text{nc}} = 0$.
- **Angular Momentum**: $\vec{L} = \vec{r} \times \vec{p}$. Conserved if net external torque is zero: $\Sigma \vec{\tau}_{\text{ext}} = \vec{0}$.

Research Skills: Error Analysis and Inquiry Labs

7.1 Introduction: Thinking Like a Scientist

Physics is not an exercise in mathematics. It is an experimental science. Every formula we have written down so far in this textbook represents a model of physical reality, and that model is only as good as the measurements that support it.

When you make a measurement in the lab, you are asking nature a question. But nature's answer is never a single, pure number. It is always a range: "The value is around here, give or take a bit." In professional science, reporting a number without its uncertainty is meaningless. If you measure the speed of light to be 3.1×10^8 m/s, is your experiment a breakthrough that disproves Einstein, or is it just a slightly noisy measurement? Without an uncertainty estimate, no one can tell.

In this chapter, we will learn how to quantify this "give or take" using the mathematics of error analysis. We will then apply these tools to a set of open, inquiry-based labs where you will design your own procedures and analyze real data.

7.2 Introduction to Error Analysis

In science, the word "error" does not mean a mistake or a blunder. It refers to the inevitable uncertainty that accompanies any physical measurement. These uncertainties fall into two categories:

1. **Random Errors**: These are unpredictable fluctuations that differ from one measurement to the next. For example, if you use a stopwatch to time a pendulum, your reaction time will vary slightly each time you click the button. Random errors can be reduced by taking multiple measurements and averaging them.
2. **Systematic Errors**: These are constant biases that skew all measurements in the same direction. For example, if your stopwatch runs 2% slow, or if your ruler is slightly warped, every measurement will be systematically off. Systematic errors cannot be detected by repeating the measurement; you must carefully calibrate your instruments.

Mean and Standard Deviation

If we repeat a measurement of a quantity x several times under identical conditions, we get a set of values: $x_1, x_2, \dots, x_N$. Our best estimate for the true value of x is the

mean (average), $\bar{x}$:

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i. \quad (108)$$

To describe how widely the measurements spread around this average, we compute the **standard deviation**, σ_x :

$$\sigma_x = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}. \quad (109)$$

The standard deviation measures the precision of a single trial. If the errors are normally distributed (following a bell curve), then about 68% of your measurements will lie within $\pm\sigma_x$ of the mean, and 95% will lie within $\pm 2\sigma_x$.

Standard Deviation of the Mean

If you repeat the entire experiment (taking another set of N measurements), the new mean will be slightly different. How certain are we of the mean itself? The uncertainty of our average is called the **standard deviation of the mean** (or the **standard error**), $\sigma_{\bar{x}}$:

$$\sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{N}}. \quad (110)$$

This is a beautiful result: it says that the more measurements you take, the more certain you become of the average. If you want to cut your uncertainty in half, you need to take $2^2 = 4$ times as many measurements. We write our final measurement as:

$$(\text{measured value}) = \bar{x} \pm \sigma_{\bar{x}}. \quad (111)$$

Propagation of Uncertainty

Often, you cannot measure the quantity you want directly. Instead, you measure other quantities and calculate the final result. For example, to find the speed v , you measure distance d and time t , and compute $v = d/t$. If d and t have uncertainties δd and δt , how do they combine to give the uncertainty δv ?

Here are the two fundamental rules of error propagation (derived from calculus):

Error Propagation Rules

Let x and y be independent measurements with uncertainties δx and δy .

- Sums and Differences**: If $z = x + y$ or $z = x - y$, the absolute uncertainties add in quadrature:

$$\delta z = \sqrt{(\delta x)^2 + (\delta y)^2}. \quad (112)$$

- Products and Quotients**: If $z = xy$ or $z = x/y$, the fractional uncertainties add in quadrature:

$$\frac{\delta z}{|z|} = \sqrt{\left(\frac{\delta x}{x}\right)^2 + \left(\frac{\delta y}{y}\right)^2}. \quad (113)$$

Worked Example Uncertainty of Speed

A student measures the distance a cart travels to be $d = 2.00 \pm 0.02$ m, and the time taken to be $t = 4.0 \pm 0.2$ s. What is the calculated speed of the cart and its uncertainty?

Solution: The nominal speed is:

$$v = \frac{d}{t} = \frac{2.00 \text{ m}}{4.0 \text{ s}} = 0.50 \text{ m/s}.$$

Since v is a quotient, we use the product/quotient rule for fractional uncertainties:

$$\frac{\delta d}{d} = \frac{0.02}{2.00} = 0.01 \quad (1\%), \quad \frac{\delta t}{t} = \frac{0.2}{4.0} = 0.05 \quad (5\%).$$

The fractional uncertainty of the speed is:

$$\frac{\delta v}{v} = \sqrt{(0.01)^2 + (0.05)^2} = \sqrt{0.0001 + 0.0025} \approx 0.051 \quad (5.1\%).$$

The absolute uncertainty is:

$$\delta v = v \times 0.051 = (0.50 \text{ m/s})(0.051) \approx 0.03 \text{ m/s}.$$

So we report:

$$v = 0.50 \pm 0.03 \text{ m/s}.$$

7.3 Inquiry-Based Lab Manual

In these labs, we do not give you a checklist of step-by-step instructions. Instead, we give you a goal, a list of available apparatus, and the physical constraints. It is your job as a group to design the experiment, identify the variables, estimate the uncertainties, and write a report defending your conclusions.

Our lab is equipped with standard materials, smartphones (which contain slow-motion video, audio recorders, and accelerometers), PASCO scientific interface equipment (including photogates, motion sensors, and force probes), and Thymio educational robots.

Lab 1: The Speed of Sound via Time-of-Flight

- ****Guiding Question**:** What is the speed of sound in the air around the Dzong courtyard, and does it agree with the theoretical formula $v = 331.3 + 0.606 T_C$ (where T_C is temperature in Celsius)?
- ****Available Apparatus**:** A long measuring tape, temperature sensor, and smartphones running an audio-recording app or a specialized acoustics app (like Phyx) that can trigger a timer on loud claps.
- ****Guided Autonomy**:** You must decide on the distance between the clap source and the receiver, how to account for human reaction times (or how to eliminate

them using two smartphones separated by distance), and how many trials are needed to make the uncertainty smaller than the temperature dependence.

Lab 2: Measuring g with a Pendulum and Free Fall

- ****Guiding Question****: What is the local value of the acceleration due to gravity g in Paro?
- ****Available Apparatus****: Strings, masses, a stand, a ruler, PASCO photogates, and PASCO motion sensors.
- ****Guided Autonomy****: You should use two different methods to determine g . Method A: measure the period of a simple pendulum as a function of length. Method B: drop an object and record its position versus time using a PASCO motion sensor or photogate. Compare the results. Which method has a smaller systematic error? Which has a smaller random error? How do you account for air resistance?

Lab 3: Projectile Launch and Range Prediction

- ****Guiding Question****: Can we predict exactly where a projectile launched at an arbitrary angle will land, including uncertainty?
- ****Available Apparatus****: A spring-loaded projectile launcher, a target plate, carbon paper to mark the landing spot, tape, a ruler, and PASCO photogates to measure launch velocity.
- ****Guided Autonomy****: First, launch the projectile horizontally multiple times to determine the launch speed v_0 (using photogates to find the mean and standard deviation). Then, calculate the predicted landing spot for a launch angle of 35° from a given height. Propagate the uncertainties of v_0 , angle, and height to draw a “target zone” on the floor. Place your target plate on the floor. Your grade is based on whether the projectile lands within your predicted uncertainty zone on the first shot!

Lab 4: Thymio Mars Rover Navigation and Noise Characterization

- ****Guiding Question****: How does measurement noise affect a robot’s ability to navigate autonomously, and how can we characterize its sensors?
- ****Available Apparatus****: A Thymio II educational robot, a laptop for programming (Aseba/Scratch/Python), distance sensor calibration targets, and obstacle courses.
- ****Guided Autonomy****: The Thymio robot uses infrared sensors to detect objects. These sensors do not report distance directly; they report a raw intensity value.
1. First, characterize the horizontal distance sensor: record the sensor reading at distances from 2 cm to 20 cm (in 2 cm increments) using a ruler. Plot this curve to find the function mapping raw readings to distance.
2. Identify the noise: record 50 readings at a fixed distance of 10 cm. Calculate the mean and standard

deviation of the raw sensor noise. 3. Design a control algorithm for a “Mars Rover” that allows the robot to drive toward a wall and stop exactly 10 ± 1 cm away. Program the robot, run 10 trials, and verify if the final stopping distance matches your target within your characterized uncertainty limits.

Whiteboard Challenge 7.1: Classifying Errors *(stand up, groups of three, one marker each)*

A student group measures the acceleration of a cart rolling down an inclined plane. In their report, they list the following issues. Classify each as a **random error**, a **systematic error**, or a **blunder** (mistake):

- (a) The track was slightly sloped sideways, causing the cart to rub against the guard rail in every trial.
- (b) The stopwatch was clicked by a student who was looking at the cart from a sharp angle, causing parallax in their timing.
- (c) The student recording the numbers wrote down 0.35 instead of 0.53 for Trial 4.
- (d) Air currents in the room caused the cart’s travel times to vary by a few hundredths of a second.

Electricity and Magnetism

8.1 Introduction: The Unified Force

For thousands of years, electricity and magnetism were seen as two completely unrelated phenomena. Electricity was about lightning, electric eels, and amber rods attracting lint when rubbed. Magnetism was about lodestones—mysterious rocks found in Thessaly that could attract iron and align themselves north-south.

In the nineteenth century, a series of breakthrough experiments revealed that they are actually two sides of the same coin: **electromagnetism**. Today, we know that static electric charges create electric fields, but when those charges start moving (forming an electric current), they create magnetic fields. In this chapter, we will build the bridge between the two, moving from static charges, through electric circuits, to magnetic forces and moving charges.

8.2 Electrostatics: Charges and Fields

The basic building block of electricity is **electric charge**, denoted by q . In the universe as we observe it on a macroscopic level, charge is always conserved and always comes in integer multiples of the elementary charge $e \approx 1.602 \times 10^{-19} \text{ C}$ (coulombs).

However, modern particle physics has revealed that the protons and neutrons inside the nucleus are not fundamental. They are made of **quarks**, which carry fractional electric charges: the up quark (u) carries $+\frac{2}{3}e$, and the down quark (d) carries $-\frac{1}{3}e$. A proton (uud) has charge $+1e$, while a neutron (udd) has charge $0e$. Because quarks are confined inside hadrons and cannot be isolated individually, the free charges we measure in the lab are still always integer multiples of e .

Historical Experiments: Coulomb and Millikan

How do charges interact, and how do we know they are quantized?

1. **Coulomb's Torsion Balance (1785)**: The French physicist Charles-Augustin de Coulomb measured the force between small charged spheres using a torsion balance—a fine silver wire holding a horizontal needle. By twisting the wire, he could measure incredibly tiny forces, showing that the electrostatic force between two point charges q_1 and q_2 separated by distance r is:

$$F_e = k_e \frac{|q_1 q_2|}{r^2}, \quad \text{where } k_e \equiv \frac{1}{4\pi\epsilon_0} \approx 8.988 \times 10^9 \text{ N m}^2/\text{C}^2. \quad (114)$$

This is **Coulomb's Law**, which looks mathematically identical to Newton's law of

gravity, but with one massive difference: gravity only attracts, while charges can repel (like charges repel, opposites attract).

2. **Millikan's Oil Drop Experiment (1909)**: The American physicist Robert A. Millikan measured the charge of the electron directly. He sprayed tiny, charged droplets of oil into a chamber between two horizontal metal plates. By adjusting the electric field between the plates, he could balance the upward electric force (qE) against the downward force of gravity (mg) and air resistance. By measuring the terminal velocity of the drops as they fell or rose, he discovered that the charge on every drop was always an integer multiple of a fundamental value: $e \approx 1.6 \times 10^{-19} \text{ C}$.

The Electric Field

How does charge q_1 push on charge q_2 across empty space? We say that q_1 sets up an **electric field** $\vec{E}$ in the surrounding space, and that field exerts a force on any test charge q_0 placed inside it:

$$\vec{E} \equiv \frac{\vec{F}_e}{q_0}. \quad (115)$$

The electric field of a single point charge q is:

$$\vec{E} = k_e \frac{q}{r^2} \hat{r}. \quad (116)$$

Electric fields are represented by field lines that emerge from positive charges and terminate on negative charges. The density of lines represents the strength of the field.

Electric Potential and Potential Difference

Because the electrostatic force is conservative, we can define an **electric potential energy** U_e . We define the **electric potential** V as the potential energy per unit charge:

$$V \equiv \frac{U_e}{q}. \quad (117)$$

The unit of potential is the joule per coulomb, renamed the **volt** (V). If a charge q moves through a potential difference ΔV , its potential energy changes by $\Delta U_e = q\Delta V$.

8.3 Electric Circuits

If we connect two points at different potentials with a conductor (like a copper wire), charges will flow from the higher potential to the lower potential. This flow of charge is an **electric current**, I :

$$I \equiv \frac{dq}{dt}. \quad (118)$$

The unit of current is the ampere (A, or “amp”), representing one coulomb per second. By convention, current is defined as the flow of **positive** charge, even though in metal wires it is actually negative electrons flowing in the opposite direction.

To maintain a continuous current, we need a device that pumps charges back to the higher potential. Such a device is a source of **electromotive force (EMF)**, denoted by $\mathcal{E}$, such as a battery or solar cell.

Ohm's Law, Resistance, and Resistivity

For most materials, the current flowing through a conductor is directly proportional to the potential difference (voltage) across it. This is **Ohm's Law**:

$$V = IR, \quad (119)$$

where R is the electrical **resistance**, measured in ohms (Ω).

The resistance of a wire depends on its geometry and the material it is made of:

$$R = \rho \frac{L}{A}, \quad (120)$$

where L is the length, A is the cross-sectional area, and ρ is the **resistivity** of the material (measured in $\Omega \cdot \text{m}$).

Resistor Combinations and Internal Resistance

Real batteries are not perfect; they have internal resistance r . The terminal voltage V delivered to a circuit is less than the EMF $\mathcal{E}$ when current flows:

$$V = \mathcal{E} - Ir. \quad (121)$$

When combining multiple resistors: - **In Series**: The current through each is the same. The equivalent resistance is the sum:

$$R_{\text{eq}} = R_1 + R_2 + R_3 + \dots \quad (122)$$

- **In Parallel**: The voltage across each is the same. The reciprocal of the equivalent resistance is the sum of the reciprocals:

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots \quad (123)$$

Electrical Power and Energy

When a current I passes through a resistor with a voltage drop V , electrical energy is converted into thermal energy. The rate of this energy transfer is the **power** P :

$$P = IV = I^2 R = \frac{V^2}{R}. \quad (124)$$

Worked Example Internal Resistance of a Car Battery

A car battery has an EMF of $\mathcal{E} = 12.6 \text{ V}$ and an internal resistance of $r = 0.08 \Omega$.

- (a) What is the terminal voltage of the battery when it is connected to a headlight of resistance $R = 5.00 \Omega$?
- (b) What is the terminal voltage when the starter motor is engaged, drawing a massive current of $I = 100 \text{ A}$?

Solution:

- (a) The total resistance of the loop is $R_{\text{total}} = R + r = 5.00 + 0.08 = 5.08 \Omega$. The current is:

$$I = \frac{\mathcal{E}}{R_{\text{total}}} = \frac{12.6 \text{ V}}{5.08 \Omega} \approx 2.48 \text{ A}.$$

The terminal voltage is:

$$V = \mathcal{E} - Ir = 12.6 - (2.48)(0.08) \approx 12.4 \text{ V}.$$

- (b) When the starter draws $I = 100 \text{ A}$:

$$V = \mathcal{E} - Ir = 12.6 - (100)(0.08) = 12.6 - 8.0 = 4.6 \text{ V}.$$

The terminal voltage drops drastically because of the internal resistance, which is why car cabin lights dim when you start the engine.

8.4 Magnetostatics: Fields and Currents

A permanent magnet has north and south poles. Like poles repel, opposite poles attract. If you cut a magnet in half, you do not get an isolated north pole and south pole; instead, you get two smaller magnets, each with its own north and south pole. There are no isolated magnetic charges (magnetic monopoles) in nature.

At the atomic level, magnetism arises from moving electric charges. Electrons orbit the nucleus and spin on their axes; each electron acts as a tiny current loop, forming a miniature **magnetic dipole**. In most materials, these dipoles point in random directions and cancel out. In ferromagnetic materials (like iron), they align themselves in domains, creating a macroscopic magnet.

Oersted's Compass Needle (1820)

During a lecture at the University of Copenhagen in April 1820, the Danish physicist Hans Christian Oersted noticed that a magnetic needle deflected whenever he switched on an electric current in a nearby wire. This was the first experimental proof that an electric current creates a magnetic field.

Shortly after, André-Marie Ampère and others worked out the rules: a current-carrying wire is surrounded by circular magnetic field lines $\vec{B}$, whose direction is given by the **right-hand rule** (point your thumb in the direction of the current, and your fingers curl in the direction of the magnetic field).

Magnetic Force on Moving Charges

While static charges feel no force from a magnetic field, a charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ experiences a force $\vec{F}_B$ given by the cross product:

Magnetic Lorentz Force

The magnetic force $\vec{F}_B$ on a charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ is:

$$\vec{F}_B = q(\vec{v} \times \vec{B}). \quad (125)$$

Its magnitude is $F_B = |q|vB \sin \theta$, where θ is the angle between $\vec{v}$ and $\vec{B}$. The force is perpendicular to both $\vec{v}$ and $\vec{B}$.

Because $\vec{F}_B$ is always perpendicular to $\vec{v}$, the work done on the charge by the magnetic field is zero:

$$P = \vec{F}_B \cdot \vec{v} = q(\vec{v} \times \vec{B}) \cdot \vec{v} = 0. \quad (126)$$

A magnetic field can change the *direction* of a moving charge's velocity, but it can never change its *speed* or its kinetic energy!

A Curved Trajectory: Uniform Circular Motion

Consider a particle of mass m and charge q entering a uniform magnetic field $\vec{B}$ perpendicular to its velocity $\vec{v}$ (Fig. 8.1).

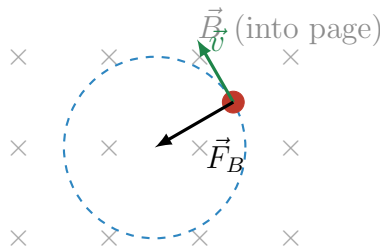


Figure 8.1: A positive charge moving perpendicular to a uniform magnetic field $\vec{B}$ (pointing into the page) undergoes uniform circular motion.

Since the magnetic force is perpendicular to the velocity, it acts as a centripetal force:

$$qvB = m \frac{v^2}{R} \implies R = \frac{mv}{qB}. \quad (127)$$

The radius of the circular path is proportional to the momentum mv and inversely proportional to the field strength B . This is the basis of mass spectrometers and particle accelerators.

Checkpoint 8.1

An electron (charge $-e$) and a proton (charge $+e$) enter a uniform magnetic field with the same velocity. The magnetic force on the proton is:

- A. In the same direction as on the electron.
- B. In the opposite direction to the electron.
- C. Zero, because protons are inside nuclei.
- D. Double the force on the electron.

B: Because the charge of the proton has the opposite sign to the electron ($q_p = +e$), the cross product direction flips, so the forces are in opposite directions.

Whiteboard Challenge 8.1: Designing a Velocity Selector *(stand up, groups of three, one marker each)*

Imagine a beam of ions containing both fast and slow particles. We want to design a device that lets only particles of a single chosen speed v pass through in a straight line.

- (a) If we apply a downward electric field $\vec{E}$, what force does a positive ion feel?
- (b) If we also apply a magnetic field $\vec{B}$ perpendicular to the velocity, how must we orient it so the magnetic force opposes the electric force?
- (c) Set the net force to zero and solve for the speed v that passes undeflected. Does this speed depend on the mass or charge of the ion?

(You should find $v = E/B$. Mass and charge cancel out, meaning this device acts as a pure velocity filter!)

Optics: Waves and Rays

9.1 Introduction: The Dual Face of Light

What is light? This question has split the physics community for centuries. Isaac Newton famously argued that light is a stream of tiny particles (“corpuscles”), which explained why light travels in straight lines and casts sharp shadows. But his contemporary Christiaan Huygens argued that light is a wave, propagating through a medium much like sound waves in air.

For a hundred years, Newton’s prestige kept the wave theory in the shadows. But in 1801, a simple experiment involving a card with two thin slits changed everything, proving beyond doubt that light can interfere with itself—a property unique to waves. Later, James Clerk Maxwell showed that light is actually an electromagnetic wave, moving at a speed dictated by the electrical and magnetic constants of space.

Yet, when light interacts with objects much larger than its wavelength (like lenses, mirrors, or water drops), we do not need to worry about wave crests and interference. We can treat light as a set of straight lines called **rays**. In this chapter, we explore both descriptions, showing how the wave description leads to ray behaviors under the beautiful unifying rule of Fermat’s principle of least time.

9.2 Wave Optics

A wave is a disturbance that travels through space, carrying energy without transporting matter. Waves can be **transverse** (where the medium oscillates perpendicular to the direction of wave travel, like waves on a string) or **longitudinal** (where the medium oscillates parallel, like sound waves).

Mathematically, a one-dimensional wave traveling along the x -axis with speed v is represented by a wavefunction $\psi(x, t)$:

$$\psi(x, t) = f(x \mp vt), \quad (128)$$

where the minus sign is for a wave traveling to the right, and the plus sign is for a wave traveling to the left. Any function of this form satisfies the **wave equation**:

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}. \quad (129)$$

Electromagnetic Waves and the Speed of Light

In 1864, Maxwell compiled the laws of electricity and magnetism and discovered that in vacuum, the electric and magnetic fields satisfy the wave equation:

$$\frac{\partial^2 E}{\partial x^2} = \mu_0 \varepsilon_0 \frac{\partial^2 E}{\partial t^2}. \quad (130)$$

Comparing this to the standard wave equation, he realized that electromagnetic disturbances must travel at speed:

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}. \quad (131)$$

Using the values of $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$ and $\varepsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2)$, he calculated:

$$c = \frac{1}{\sqrt{(4\pi \times 10^{-7})(8.854 \times 10^{-12})}} \approx 2.998 \times 10^8 \text{ m/s}. \quad (132)$$

This matched the measured speed of light exactly! Maxwell wrote: “we can scarcely avoid the inference that light consists in the transverse undulations of the same medium which is the cause of electric and magnetic phenomena.”

Rayleigh Scattering: Why the Sky is Blue

When light hits small molecules in the atmosphere, it scatters. The efficiency of this scattering is inversely proportional to the fourth power of the wavelength:

$$I_{\text{scattered}} \propto \frac{1}{\lambda^4}. \quad (133)$$

Because blue light has a much shorter wavelength than red light ($\lambda_{\text{blue}} \approx 400 \text{ nm}$ vs $\lambda_{\text{red}} \approx 700 \text{ nm}$), blue light is scattered about ten times more efficiently. When we look up, we see this scattered blue light. At sunset, the light must travel through a much thicker layer of atmosphere, which scatters away all the blue, leaving only the direct red and orange wavelengths to reach our eyes.

Young’s Double-Slit Experiment

In 1801, Thomas Young split a thin beam of sunlight using two closely spaced slits. On a screen behind them, he saw not two bright lines, but a series of bright and dark bands (fringes).

This interference is explained by the path length difference ΔL between the waves from the two slits. If the slits are separated by distance d and we look at angle θ :

$$\Delta L = d \sin \theta. \quad (134)$$

- **Constructive Interference (Bright Fringes)**: If the path difference is an integer number of wavelengths, the waves arrive in phase:

$$d \sin \theta = m\lambda, \quad m = 0, \pm 1, \pm 2, \dots \quad (135)$$

- **Destructive Interference (Dark Fringes)**: If the path difference is a half-integer, they arrive out of phase and cancel out:

$$d \sin \theta = \left(m + \frac{1}{2}\right) \lambda, \quad m = 0, \pm 1, \pm 2, \dots \quad (136)$$

9.3 Ray Optics

When light hits a boundary between two media, it is partially **reflected** and partially **refracted** (bent). Lenses and mirrors work by bending rays of light.

The bending of light is characterized by the **index of refraction** n of the material, defined as the ratio of light speed in vacuum to its speed in the medium:

$$n \equiv \frac{c}{v}. \quad (137)$$

Air has $n \approx 1.0$, water has $n \approx 1.33$, and glass has $n \approx 1.5$.

Fermat's Principle of Least Time

In 1662, the French mathematician Pierre de Fermat proposed a beautiful principle that generates all of ray optics:

Fermat's Principle of Least Time

A ray of light traveling between two points takes the path that requires the **least time** compared to nearby paths.

Let us use Fermat's principle to derive **Snell's Law of Refraction**.

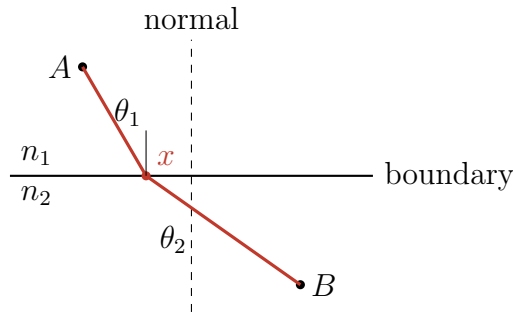


Figure 9.1: Light travels from A to B , crossing the boundary at position x . The path of least time bends at the interface.

Let point A be at coordinates $(0, y_1)$ in medium 1 (n_1), and point B be at $(d, -y_2)$ in medium 2 (n_2). The light crosses the boundary at $(x, 0)$ (Fig. 9.1). The total time taken is:

$$t(x) = \frac{d_1}{v_1} + \frac{d_2}{v_2} = \frac{n_1}{c} \sqrt{x^2 + y_1^2} + \frac{n_2}{c} \sqrt{(d-x)^2 + y_2^2}. \quad (138)$$

To minimize this time, we set the derivative $dt/dx = 0$:

$$\frac{dt}{dx} = \frac{n_1}{c} \frac{x}{\sqrt{x^2 + y_1^2}} - \frac{n_2}{c} \frac{d-x}{\sqrt{(d-x)^2 + y_2^2}} = 0. \quad (139)$$

From the geometry in Fig. 9.1, the terms inside are exactly the sines of the angles θ_1 and θ_2 :

$$\sin \theta_1 = \frac{x}{\sqrt{x^2 + y_1^2}}, \quad \sin \theta_2 = \frac{d-x}{\sqrt{(d-x)^2 + y_2^2}}. \quad (140)$$

Substituting these:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2. \quad (141)$$

This is **Snell's Law**! The refraction of light is simply the consequence of light taking the fastest possible path across a boundary where its speed changes.

Checkpoint 9.1

Light travels from water ($n = 1.33$) to glass ($n = 1.50$). The refracted ray:

- A. Bends toward the normal. B. Bends away from the normal.
- C. Does not bend. D. Is totally internally reflected.

A. Since $n_2 > n_1$, the speed of light is lower in glass, so the angle must decrease ($\theta_2 < \theta_1$) to maintain the equality $n_1 \sin \theta_1 = n_2 \sin \theta_2$. The ray bends toward the normal.

Supplementary — The Physics of Rainbows *(optional — read for pleasure, skip without loss)*

A rainbow is one of nature's most spectacular displays of optics, combining refraction, reflection, and dispersion inside falling raindrops.

When sunlight hits a spherical raindrop, it is refracted at the front surface, reflected off the back surface, and refracted again as it exits the front-bottom of the drop. Because water has **dispersion** (its index of refraction depends slightly on color: $n_{\text{violet}} \approx 1.344$ and $n_{\text{red}} \approx 1.331$), different colors bend at slightly different angles:

- Red light exits at a maximum concentration angle of about 42° relative to the incoming sunlight.
- Violet light exits at a maximum concentration angle of about 40° .

If you calculate the deflection angle $\theta(b)$ as a function of the impact parameter b (where the ray hits the drop relative to the center line), you find that the deflection angle reaches a maximum at a specific impact parameter, creating a concentration of rays (a "rainbow ray") at 42° . This is why the rainbow is a circle of radius 42° centered on your anti-solar point (the shadow of your head). Since violet bends more, it forms the inner edge of the primary bow, while red forms the outer edge.

If light undergoes **two** internal reflections inside the raindrop before exiting, it forms the fainter **secondary rainbow** at an angle of 51° . Because of the extra reflection, the colors of the secondary rainbow are reversed, with red on the inside and violet on the outer edge.

Whiteboard Challenge 9.1: Total Internal Reflection *(stand up, groups of three, one marker each)*

When light travels from a dense medium (like water, $n_1 = 1.33$) to a less dense medium (like air, $n_2 = 1.0$), it bends away from the normal.

- (a) What happens to the refracted angle θ_2 as you increase the incident angle θ_1 ?
- (b) Calculate the critical angle θ_c at which the refracted ray would theoretically bend to 90° (skimming the surface).
- (c) What happens if the incident angle is greater than θ_c ? As a group, decide: does any light escape into the air?

(You should find $\theta_c = \sin^{-1}(1/1.33) \approx 48.8^\circ$. For angles larger than this, no light is refracted; it is 100% reflected back into the water, a phenomenon called total internal reflection, which is how fibre-optic cables transmit data.)

Modern Physics: A Quantum Preview

10.1 Introduction: The Crack in the Classical World

By the late nineteenth century, physicists believed they had nearly finished their work. Mechanics had been solved by Newton, thermodynamics by Joule and Boltzmann, and electromagnetism by Maxwell. The universe seemed to run like a clockwork machine: if you knew the positions and velocities of all particles at one instant, you could predict their future for all eternity.

But as scientists began probing the subatomic scale, this beautiful clockwork picture fell apart. Classical physics could not explain why hot objects glowed with the colors they did, or why light shining on metal could knock out electrons in ways that defied wave theory. Solving these puzzles required a radical rewrite of the laws of nature: **quantum mechanics**. In this final chapter, we take a preview look at this strange subatomic world where particles act like waves, waves act like particles, and probability replaces certainty.

10.2 The Photoelectric Effect

In 1887, Heinrich Hertz noticed that electric sparks jumped more easily between electrodes when they were illuminated with ultraviolet light. This phenomenon—where light shining on a metal surface ejects electrons—is the **photoelectric effect**.

According to classical electromagnetic wave theory, light is a continuous wave carrying energy. The electric field of the wave oscillates and shakes the electrons in the metal. If you turn up the light intensity, the electric field becomes stronger, which should shake the electrons harder and eject them with higher kinetic energy. Furthermore, if the light is very dim, it should take time (minutes or hours) for an electron to absorb enough energy to break free.

But experiments showed the exact opposite: 1. **No Time Delay**: Even in extremely dim light, electrons are ejected almost instantly. 2. **Frequency Threshold**: If the frequency of the light is below a certain threshold f_0 , no electrons are ejected, no matter how intense the light is. 3. **Intensity Rules Number, Not Energy**: Turning up the intensity increases the **number** of ejected electrons per second, but does not change their maximum kinetic energy. That energy depends only on the **frequency** of the light.

Einstein's Photon Hypothesis (1905)

Albert Einstein solved the mystery by proposing that light is not a continuous wave, but is quantized into tiny packets of energy called **photons**.

Energy of a Photon

The energy E of a single photon is directly proportional to its frequency f :

$$E = hf, \quad (142)$$

where $h \approx 6.626 \times 10^{-34} \text{ J s}$ is Planck's constant.

When a photon strikes the metal, it is absorbed by a single electron in an all-or-nothing collision. To escape the metal, the electron must overcome an attractive potential energy barrier called the **work function**, Φ . Any remaining energy becomes the electron's maximum kinetic energy, K_{max} :

$$K_{\text{max}} = hf - \Phi. \quad (143)$$

This formula explained all the experimental observations in one stroke. It won Einstein the 1921 Nobel Prize in Physics (not his work on relativity, which was still considered too speculative at the time!).

10.3 Wave-Particle Duality

If light—which we proved was a wave in Chapter 9—can act like a stream of particles (photons), can objects we think of as solid particles (like electrons) act like waves?

In 1924, a French graduate student named Louis de Broglie proposed exactly that. He argued that if nature is symmetric, then matter must also have a wave nature.

The de Broglie Relation

A particle with momentum $p = mv$ has an associated wavelength λ :

$$\lambda = \frac{h}{p}. \quad (144)$$

For macroscopic objects, Planck's constant is so tiny that the wavelength is completely undetectable. A thrown cricket ball has a wavelength of about 10^{-34} m —far too small to interact with any obstacle. But for an electron moving at high speed, its wavelength is around 10^{-10} m , which is the size of the spacing between atoms in a crystal.

Historical Experiment: Davisson-Germer (1927)

Clinton Davisson and Lester Germer fired a beam of electrons at a nickel crystal in a vacuum. They discovered that the electrons did not scatter randomly; instead, they formed a distinct diffraction pattern of peaks and valleys, exactly like X-rays reflecting off a crystal. The atomic grid of the nickel acted as a diffraction grating, proving that electrons are waves.

The Quantum Mystery: The Double-Slit with Electrons

To see the weirdness of quantum mechanics, let us compare three double-slit experiments:

1. **With Bullets**: If we fire bullets at a double-slit, they pass through one slit or the other. We get a simple pile of bullets behind each slit. The distribution is $I_{\text{both}} = I_1 + I_2$.
2. **With Waves**: If we block one slit, we get a single peak. If we open both, the waves interfere, creating an interference pattern with bright and dark spots: $I_{\text{both}} \neq I_1 + I_2$.
3. **With Electrons**: If we fire electrons *one by one*, each electron lands as a single dot on the detector (particle behavior). We cannot predict where any single electron will land. But as we fire thousands of electrons one by one, the dots gradually build up to form a perfect double-slit interference pattern (wave behavior)!

Each electron seems to pass through *both* slits simultaneously, interfering with itself, before collapsing to a single point on the screen.

10.4 Foundations of Quantum Mechanics

How do we describe this behavior mathematically? In 1926, Erwin Schrödinger wrote down the wave equation for matter, whose solution is the **wavefunction** $\Psi(x, t)$.

Unlike a water wave where the amplitude is the physical height, the wavefunction itself is not directly measurable. Instead, the German physicist Max Born proposed that the square of the wavefunction represents a probability:

Born's Probability Interpretation

The probability $P(x) dx$ of finding a particle in a small range dx around position x is:

$$P(x) dx = |\Psi(x, t)|^2 dx. \quad (145)$$

In the quantum world, determinism is gone. We cannot calculate where a particle *is*; we can only calculate the *probability* of finding it there.

Watching the Electrons: Measurement Collapse

What if we place a light source behind the double-slit to watch which slit each electron goes through? As soon as we do this, the measurement forces each electron to choose a path. The instant we detect which slit the electron passed through, the interference pattern disappears! The act of measurement collapses the quantum wave state, returning the system to classical, bullet-like behavior.

The Heisenberg Uncertainty Principle

Because a particle is described by a wave packet, we cannot specify both its position and its momentum with absolute certainty. Werner Heisenberg proved that the product of their uncertainties has a fundamental limit:

Heisenberg Uncertainty Principle

$$\Delta x \Delta p \geq \frac{\hbar}{2}, \quad \text{where } \hbar \equiv \frac{h}{2\pi}. \quad (146)$$

If you design an experiment to pin down the position of an electron ($\Delta x \rightarrow 0$), its momentum becomes completely wild and unpredictable ($\Delta p \rightarrow \infty$). This is not a limitation of our instruments; it is a fundamental property of wave-like matter.

Checkpoint 10.1

If the wavelength of a light wave is cut in half, the energy of its photons is:

- A. Cut in half. B. Doubled.
C. Quadrupled. D. Unchanged.

B. Since $f = c/\lambda$, cutting the wavelength in half doubles the frequency ($f \rightarrow 2f$), which doubles the photon energy $E = hf$.

Whiteboard Challenge 10.1: Rutherford's Gold Foil *(stand up, groups of three, one marker each)*

In 1911, Ernest Rutherford's students fired alpha particles (heavy positive helium nuclei) at a thin sheet of gold foil.

- (a) If the gold atoms were like J.J. Thomson's "plum pudding" model (a soft sphere of positive charge with tiny electrons scattered inside), what should happen to the alpha particles?
- (b) In the actual experiment, most passed straight through, but about 1 in 8000 bounced almost straight back. Rutherford wrote: "It was as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you." What does this say about where the mass and positive charge of the atom are concentrated?

(It proved that the atom is mostly empty space, with almost all its mass and positive charge concentrated in a tiny, dense core called the nucleus.)